



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru- 560 078.

CURRICULUM

For the Academic year 2021-25

DEPARTMENT OF ARTIFICIAL INTELLIGENCE &

MACHINE LEARNING

**2021 Scheme
III to VIII Semester B.E
2021 – 2025 Batch**



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About the Institute

Founded in 1979, DSCE is a premier engineering college in India, renowned for its academic excellence and industry-oriented programs. With state-of-the-art infrastructure, dedicated faculty, and a strong focus on research and innovation, DSCE has consistently ranked high in national rankings. The college offers a wide range of undergraduate and postgraduate programs, preparing students for successful careers in the engineering field.

DSCE has established strong partnerships with leading multinational corporations, creating Centres of Excellence to bridge the gap between academia and industry. These partnerships provide students with valuable practical experience and exposure to real-world challenges. Additionally, the college's alumni association plays a vital role in connecting graduates with the institution and providing opportunities for networking and professional development.

The college's commitment to fostering innovation and entrepreneurship is evident through its various initiatives, such as the Innovation and Entrepreneurship Development Centre (IEDC) and the Centre for Innovation & Leadership (CIL). These centers provide students with the necessary resources and support to develop their entrepreneurial skills and become successful leaders.

DSCE's commitment to holistic development is evident in its focus on extracurricular activities and student well-being. The college offers a variety of clubs and societies that cater to students' interests and hobbies, fostering a vibrant campus culture. Additionally, DSCE prioritizes student health and well-being, providing state-of-the-art sports facilities and counseling services.

DSCE's legacy of excellence is reflected in the achievements of its alumni, who have gone on to hold influential positions in various industries. The college's strong alumni network provides valuable mentorship and career opportunities for current students. DSCE continues to evolve and adapt to the changing landscape of higher education, ensuring that its graduates are equipped with the skills and knowledge necessary to thrive in the competitive global market.

About the Department

The Department of AI & ML at DSCE is an emerging hub of expertise in cutting-edge technology. With a comprehensive curriculum, qualified faculty, and state-of-the-art resources, the department provides students with a strong foundation in artificial intelligence and machine learning.

The highly experienced faculty brings a wealth of subject expertise to the classroom. With extensive practical knowledge and academic backgrounds, they serve as dedicated mentors who inspire students to excel and prepare them for successful careers.

The faculty's diverse specializations lead to multi-disciplinary research projects in various domains. This promotes students to participate actively in research and publish papers in reputed journals and conferences.

The UG program's comprehensive range of courses covers various AI & ML domains, such as Deep Learning, Machine Learning, Advanced Machine Learning, MLOps, NLP, IoT, and Cloud Computing.

The department proudly holds Memorandums of Understanding (MoUs) with a large number of esteemed partners. These strategic partnerships foster innovation, collaboration, and knowledge sharing in the industry, benefiting all stakeholders and advancing shared goals.

VISION AND MISSION OF THE INSTITUTE

VISION

To impart quality technical education with a focus on Research and Innovation emphasizing on Development of Sustainable and Inclusive Technology for the benefit of society

MISSION

- To provide an environment that enhances creativity and Innovation in pursuit of excellence.
- To nurture team work in order to transform individuals as responsible leaders and entrepreneurs.
- To train the students to the changing technical scenario and make them to understand the importance of Sustainable and Inclusive technologies.

VISION AND MISSION OF THE DEPARTMENT

VISION

To provide progressive education and flourish the student's ingenuity to be successful professionals impacting the society for a smarter and ethical world.

MISSION

- To adopt an engaging teaching learning process with emphasis on problem solving and programming skills.
- To promote additional skill development through enhanced and experiential learning.
- To collaborate with industries and professional bodies and make the students industry ready.
- To encourage innovation through multi-disciplinary research and development activities.
- To imbibe human values and ethics in students to make them impact the society's as responsible professionals.

PROGRAM SPECIFIC OBJECTIVES (PSOs)

- **PSO 1:** Graduates will be adept in programming and problem solving skills for developing and managing software.
- **PSO 2:** Graduates will be able to identify, formulate, predict and solve real world problems by applying principles of Artificial Intelligence & Machine Learning.
- **PSO 3:** Graduates will be proficient to efficiently manage computer system resources using cloud computing technologies.

PROGRAM EDUCATION OUTCOMES (PEOs)

Engineering Graduates will be able to:

- PEO 1: Able to practice and implement their success skills like problem solving, communication and collaboration for providing innovative engineering solutions.
- PEO 2: Contribute their AI & ML expertise grounded in computer science and mathematics as members and leaders of professional engineering teams in multidisciplinary domains.
- PEO 3: Demonstrate lifelong learning through continued professional development and higher education in top graduate technical, research and management programs.
- PEO 4: Demonstrate a commitment to society by applying the skills and knowledge for a smarter and ethical world.

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design / development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

SEMESTER-WISE COURSE COMPONENTS CREDIT DISTRIBUTION

COURSE CATEGORY	SEMESTER								
	I	II	III	IV	V	VI	VII	VIII	Total
Humanities and Social Sciences including Management courses, Ability Enhancement	3	3	3	5	4		3		21
Basic Sciences	8	8	3	3					22
Engineering Science courses including workshop, drawing, basics of electrical/mechanical / computer, etc.	9	9							18
Professional core courses			12	12	14	11	3		52
Professional Elective courses relevant to the chosen specialization/branch						3	5		8
Open subjects–Electives from other technical and/or emerging subjects						3	3		6
Project work, seminar, and internship in industry or appropriate workplace / academic and research institutions in India / abroad				2		5	10	16	33
TOTAL	20	20	18	22	18	22	24	16	160



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SCHEME OF TEACHING AND EXAMINATIONS (2021 Batch)

(As Per New Education Policy 2020)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

III SEMESTER

#	Course and Course Code		Course Title	Syllabus and QP Setting	Teaching Department	Teaching Hours/Week				Examination				Credits
						L	T	P	S	Duration (hrs)	CIE	SEE	Total	
1	BSC	21MAT31	Foundation in Mathematics for	MAT	MAT	3	0	0	0	03	50	50	100	3
2	IPCC	21AI32	Data Structures with Applications	AIML	AIML	3	0	2	0	03	50	50	100	4
3	IPCC	21AI33	Database Management System	AIML	AIML	3	0	2	0	03	50	50	100	4
4	PCC	21AI34	Software Engineering	AIML	AIML	2	2	0	0	03	50	50	100	3
5	PCCL	21AI35	Application Development Using Python	AIML	AIML	0	0	2	0	03	50	50	100	1
6	SCR	21SCR36	Social Connect and Responsibility	AIML	AIML	1	0	0	0	02	50	50	100	1
7	HSMC	21KSK37	Sanskritika Kannada	HSMC	HSMC	1	0	0		02	50	50	100	1
	HSMC	21KBK37	Balake Kannada											
	Or													
	HSMC	21CIP37	Constitution of India and Professional Ethics											
8	AEC	21AI38X	Ability Enhancement Course-III	AIML	AIML	Theory Course				02	50	50	100	1
						1	0	0	0					
						Lab Course				02				
						0	0	2	0					
Total											400	400	800	18

A-Mathematics Course for IT branches, B-Mathematics Course for circuit branches, C-Non-circuit Branches.

Course prescribed to lateral entry Diploma holders admitted to III semester B.E./B.Tech programs													
09	NMC	21MATDIP31	Additional Mathematics - I	Maths	02	02	0	0	---	100	---	100	0



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Ability Enhancement Course-III*			
21AIL381	Robotic Process Automation	21AIL383	
21AIL382		21AIL384	

***Note: If ability enhancement course is offered as laboratory, keep the subject code as 21XXL38X**

Note:

- 1) BSC: Basic Science Course, IPCC: Integrated Professional Core Course, PCC: Professional Core Course, INT –Internship, HSMC: Humanity and Social Science & Management Courses, AEC–Ability Enhancement Courses. UHV: Universal Human Value Course.
- 2) L –Lecture, T – Tutorial, P- Practical/ Drawing, S- Self study
- 3) 21KSK37/47 Samskrutika Kannada is for students who speak, read and write Kannada and 21KBK37/47 Balake Kannada is for non-Kannada speaking, reading, and writing students.
- 4) Integrated Professional Core Course (IPCC): Refers to Professional Theory Core Course Integrated with Practicals of the same course. The theory part of the IPCC shall be evaluated both by CIE and SEE. The practical part shall be evaluated by only CIE (no SEE). However, questions from the practical part of IPCC shall be included in the SEE question paper.

21INT410 Inter/Intra Institutional Internship:

All the students admitted to engineering programs under the lateral entry category shall have to undergo a mandatory inter/intra institutional internship of 03 weeks during the intervening period of III and IV semesters. The internship shall be slated for CIE only and will not have SEE. The letter grade earned through CIE shall be included in the IV semester grade card. The internship shall be considered as a head of passing and shall be considered for vertical progression and for the award of Degree. Those, who don't takeup/complete the internship shall be declared fail and shall have to complete during subsequently after satisfying the internship requirements. The faculty coordinator or mentor shall monitor the students internship progress and interact with them for the successful completion of the internship.

Scheduled activities for III to VI semesters			
21NS83	National Service Scheme (NSS)	NSS	All students have to register for any one of the courses namely National Service Scheme, Physical Education (PE)(Sports and Athletics) and Yoga with the concerned coordinator of the course during the first week of III semester. The activities shall be



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21PE83	Physical Education (PE)(Sports and Athletics)	PE	carried out (for 4 semesters) between III semester to VI semester. For the successful completion of the registered course, it is necessary for the student to provide a participation certificate to fulfil the requirements of this mandatory course for the award of degree. The events shall be appropriately scheduled by the colleges and the same shall be reflected in the calendar prepared for the NSS, PE and Yoga activities.
21YO83	Yoga	Yoga	

Non–Credit Mandatory Courses (NCMC):

A. Additional Mathematics I and II:

- 1) These courses are prescribed for III and IV semesters respectively to lateral entry Diploma holders admitted to III semester of B.E./B.Tech., programs. They shall attend the classes during the respective semesters to complete all the formalities of the course and appear for the Continuous Internal Evaluation (CIE). In case, any student fails to register for the said course/fails to secure the minimum 40 % of the prescribed CIE marks, he/she shall be deemed to have secured an F grade. In such a case, the student has to fulfill the course requirements during subsequent semester/s to earn the qualifying CIE marks. These courses are slated for CIE only and has no SEE.
- 2) Additional Mathematics I and II shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the courses shall be mandatory for the award of degree.
- 3) Successful completion of the courses Additional Mathematics I and II shall be indicated as satisfactory in the grade card. Non-completion of the courses Additional Mathematics I and II shall be indicated as Unsatisfactory.

B. National Service Scheme/Physical Education (Sport and Athletics)/ Yoga:

- 1) In case, any student fails to register for NSS, PE or Yoga and fails to submit the requirements, he/she shall be deemed to have not completed the requirements of the course. In such a case, the student has to fulfill the course requirements during subsequent semesters.
- 2) Successful completion of the course shall be indicated as satisfactory in the VIII semester grade card. Non-completion of the course shall be indicated as Unsatisfactory.
- 3) These courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the courses shall be mandatory for the award of degree.



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IV SEMESTER

#	Course and Course Code		Course Title	Syllabus and QP Setting	Teaching Department	Teaching Hours/Week				Examination				Credits
						L	T	P	S	Duration	CIE	SEE	Total	
1	BSC	21MAT41A	Discrete Mathematical Structures	MAT	MAT	3	0	0	0	03	50	50	100	3
2	IPCC	21AI42	Design and Analysis of Algorithms	AIML	AIML	3	0	2	0	03	50	50	100	4
3	IPCC	21AI43	Machine Learning Essentials	AIML	AIML	3	0	2	0	03	50	50	100	4
4	PCC	21AI44	Operating System	AIML	AIML	2	2	0	0	03	50	50	100	3
5	AEC	21BE45	Biology For Engineers	BT	AIML	2	0	0	0	02	50	50	100	2
6	PCC	21APL46	OOPs with Java	AIML	AIML	0	0	2	0	03	50	50	100	1
7	HSMC	21KSK47	Samskrutika Kannada	HSMC	HSMC	1	0	0	0	02	50	50	100	1
	HSMC	21KBK47	Balake Kannada											
	OR													
	HSMC	21CIP47	Constitution of India & professional ethics											
8	AEC	21AI48X	Ability Enhancement Course-IV	AIML	AIML	Theory course				02	50	50	100	1
						1	0	0	0					
						Lab Course				02				
						0	0	2	0					
9	UHV	21UH49	Universal Human Values	IM	AIML	1	0	0	0	02	50	50	100	1
10	INT	21INT410	Inter/Intra Institutional Internship*	-	Completed during the intervening period of II and III semesters				03	100	-	100	2	
Total											550	450	1000	22
A-Mathematics Course for IT branches, B-Mathematics Course for circuit branches, C-Non-circuit Branches, D-Mathematics course for BT														

A-Mathematics Course for IT branches, B-Mathematics Course for circuit branches, C-Non-circuit Branches, D-Mathematics course for BT

Course prescribed to lateral entry Diploma holders admitted to III semester B.E./B.Tech programs														
11	NCMC	21MATDIP41	ADDITIONAL MATHEMATICS - II	Maths	02	02	0	0	---	100	---	100	0	



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Ability Enhancement Course-IV*			
21AIL481	Data Visualization Lab	21AIL483	
21AIL482		21AIL484	

***Note: If ability enhancement course is offered as laboratory, keep the subject code as 21XXL48X**

Note:

Internship:

Internship should be completed during the intervening period of III and IV semesters by students admitted to first year of BE./B.Tech and during the intervening period of III and IV semesters by Lateral entry students admitted to III semester.

Internship of 04 weeks during the intervening period of IV and V semesters to be completed on Innovation/ Entrepreneurship/Societal based Internship.

- 1) All the students shall have to undergo a mandatory internship of 04 weeks during the intervening period of III and IV semesters. The internship shall be slated for CIE only and will not have SEE. The letter grade earned through CIE shall be included in the VI semester grade card. The internship shall be considered as a head of passing and shall be considered for vertical progression and for the award of degree. Those, who do not take up / complete the internship shall be considered under F (fail) grade and shall have to complete during subsequently after satisfying the internship requirements.
- 2) **Innovation/ Entrepreneurship Internship** shall be carried out at industry, State and Central Government /Non-government organizations (NGOs), micro, small and medium enterprise (MSME), Innovation centers or Incubation centers. Innovation need not be a single major breakthrough; it can also be a series of small or incremental changes. Innovation of any kind can also happen outside of the business world. Entrepreneurship internships offers a chance to gain hands on experience in the world of entrepreneurship and helps to learn what it takes to run a small entrepreneurial business by performing intern duties with an established company. This experience can then be applied to future business endeavors. Start-ups and small companies are a preferred place to learn the business tack ticks for future entrepreneurs as learning how a small business operates will serve the intern well when he/she manages his/her own company. Entrepreneurship acts as a catalyst to open the minds to creativity and innovation. Entrepreneurship internship can be from several sectors, including technology, small and medium-sized, and the service sector.
- 3) Societal or social internship. Urbanization is increasing on a global scale; and yet, half the world's population still resides in rural areas and is devoid of many things that urban population enjoy. Rural internship, is a work-based activity in which students will have a chance to solve/reduce the problems of the rural place for better living. As proposed under the AICTE rural internship programme, activities under Societal or social internship, particularly in rural areas, shall be considered for 40 points under AICTE activity point programme.



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V SEMESTER

#	Course and Course Code		Course Title	Teaching Department	Syllabus and QP Setting	Teaching Hours/Week				Examination				Credits
						L	T	P	S	Duration	CIE	SEE	Total	
1	PCC	21AI51	Artificial Intelligence	AIML	AIML	2	2	0	0	03	50	50	100	3
2	IPCC	21AI52	Advanced Machine Learning	AIML	AIML	3	0	2	0	03	50	50	100	4
3	PCC	21AI53	Computer Networks and Communication	AIML	AIML	2	2	0	0	03	50	50	100	3
4	PCC	21AI54	Introduction to Robotics	AIML	AIML	2	2	0	0	03	50	50	100	3
5	PCC	21AIL55	Computer networks Lab	AIML	AIML	0	0	2	0	03	50	50	100	1
6	AEC	21RM56	Research Methodology & Intellectual Property Rights	AIML	ME	2	0	0	0	02	50	50	100	2
7	HSMC	21ES57	Environmental Studies	AIML	CV	1	0	0	0	02	50	50	100	1
8	AEC	21AI58X	Ability Enhancement Course-V	AIML	AIML	Theory course				02	50	50	100	1
						1	0	0	0					
						Lab Course				02				
						0	0	2	0					
Total											400	400	800	18

Ability Enhancement Course-V*

21AIL581	R Programming Lab	21AIL583	Linux Programming Lab
21AIL582	Software design Lab	21AIL584	Full Stack Development Lab

***Note: If ability enhancement course is offered as laboratory, keep the subject code as 21XXL58X**

Mini Project: Mini Project is a laboratory-oriented/hands on course that will provide a platform to students to enhance their practical knowledge and skills by the development of small systems/applications etc. Based on the ability/abilities of the student/s and recommendations of the mentor, a single discipline or a multidisciplinary Mini- project can be assigned to a group having not more than 4 students. **The group of the**



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mini project should be framed and guide should be allocated at the first week of the fifth semester. The final evaluation of the Mini Project – II will be done at the end of sixth semester.

VI SEMESTER

#	Course and Course Code		Course Title	Teaching Department	Syllabus and QP Setting	Teaching Hours/Week				Examination				Credits
						L	T	P	S	Duration	CIE	SEE	Total	
1	PCC	21AI61	Natural Language Processing	AIML	AIML	2	2	0	0	03	50	50	100	3
2	IPCC	21AI62	Applied Big Data and Cloud Computing	AIML	AIML	3	0	2	0	03	50	50	100	4
3	PCC	21AI63	MLOps	AIML	AIML	2	2	0	0	03	50	50	100	3
4	PEC	21AI64X	Professional Elective Course-I	AIML	AIML	3	0	0	0	03	50	50	100	3
5	OEC	21AI65X	Open Elective Course-I	AIML	AIML	3	0	0	0	03	50	50	100	3
6	PCC	21AIL66	NLP Lab	AIML	AIML	0	0	2	0	03	50	50	100	1
7	MP	21AIMP67	Mini Project	-	-	Two contact hours /week				-	100	-	100	2
8	INT	21INT68	Innovation/Entrepreneurship/Societal Internship	-		To be completed during the intervening period of				--	100	-	100	3
Total											500	300	800	22

PROFESSIONAL ELECTIVE COURSE -I

21AI641	Cyber security	21AI643	IoT and 5G
21AI642	Block chain Technology	21AI644	Augmented and Virtual Reality

OPEN ELECTIVES – I (offered by the Department to other Department students)

21AI651	Introduction to AI	21AI652	Linux Shell Programming



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Note:

Internship:

Internship should be completed during the intervening period of IV and V semesters.

Internship of 04 weeks during the intervening period of IV and V semesters to be completed on Innovation/Entrepreneurship/ Societal based Internship.

- 1) All the students shall have to undergo a mandatory internship of 04 weeks during the intervening period of IV and V semesters. The internship shall be slated for CIE only and will not have SEE. The letter grade earned through CIE shall be included in the VI semester grade card. The internship shall be considered as a head of passing and shall be considered for vertical progression and for the award of degree. Those, who do not take up / complete the internship shall be considered under F (fail) grade and shall have to complete during subsequently after satisfying the internship requirements.
- 2) **Innovation/ Entrepreneurship Internship** shall be carried out at industry, State and Central Government /Non-government organizations (NGOs), micro, small and medium enterprise (MSME), Innovation centers or Incubation centers. Innovation need not be a single major breakthrough; it can also be a series of small or incremental changes. Innovation of any kind can also happen outside of the business world. Entrepreneurship internships offers a chance to gain hands on experience in the world of entrepreneurship and helps to learn what it takes to run a small entrepreneurial business by performing intern duties with an established company. This experience can then be applied to future business endeavors. Start-ups and small companies are a preferred place to learn the business ticks for future entrepreneurs as learning how a small business operates will serve the intern well when he/she manages his/her own company. Entrepreneurship acts as a catalyst to open the minds to creativity and innovation. Entrepreneurship internship can be from several sectors, including technology, small and medium-sized, and the service sector.
- 3) **Societal or social internship.** Urbanization is increasing on a global scale; and yet, half the world's population still resides in rural areas and is devoid of many things that urban population enjoy. Rural internship, is a work-based activity in which students will have a chance to solve/reduce the problems of the rural place for better living. As proposed under the AICTE rural internship programme, activities under Societal or social internship, particularly in rural areas, shall be considered for 40 points under AICTE activity point programme.

Professional Elective Courses (PEC): A professional elective (PEC) course is intended to enhance the depth and breadth of educational experience in the Engineering and Technology curriculum. Multidisciplinary courses that are added supplement the latest trend and advanced technology in the selected stream of engineering. Each group will provide an option to select one course. The minimum number of students' strengths for offering professional electives is 10. However, this condition shall not be applicable to cases where the admission to the program is less than 10.



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Open Elective Courses:

Students belonging to a particular stream of Engineering and Technology are not entitled to the open electives offered by their parent Department. However, they can opt for an elective offered by other Departments, provided they satisfy the prerequisite condition if any. Registration to open electives shall be documented under the guidance of the Program Coordinator/ Advisor/Mentor. The minimum numbers of students' strength for offering an Open Elective Course is 10. However, this condition shall not be applicable to class where the admission to the program is less than 10.

MINI-PROJECT:

Mini Project is a laboratory-oriented/hands on course that will provide a platform to students to enhance their practical knowledge and skills by the development of small systems/applications etc. Based on the ability/abilities of the student/s and recommendations of the mentor, a single discipline or a multidisciplinary Mini- project can be assigned to a group having not more than 4 students.

CIE procedure for Mini-project

1. **Single discipline:** The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two faculty members of the Department, one of them being the Guide. The CIE marks awarded for the Mini-project work shall be based on the evaluation of the project report, project presentation skill, and question and answer session in the ratio of 50:25:25. The marks awarded for the project report shall be the same for all the batches mates.
2. **Interdisciplinary:** Continuous Internal Evaluation shall be group-wise at the college level with the participation of all the guides of the project.

The CIE marks awarded for the Mini-project, shall be based on the evaluation of the project report, project presentation skill, and question and answer session in the ratio 50:25:25. The marks awarded for the project report shall be the same for all the batch mates.

No SEE component for Mini-Project.



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SCHEME OF TEACHING AND EXAMINATIONS (2021 Batch)

(As Per New Education Policy 2020)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

VII SEMESTER

	Course and Course Code		Course Title	Teaching Department	Syllabus and QP Setting	Teaching Hours/Week				Examination				Credits
						L	T	P	S	Duration	CIE	SEE	Total	
1	HSMC	21HS71	Management and Entrepreneurship	AIML	ME	3	0	0	0	03	50	50	100	3
2	PCC	21AI72	Generative AI	AIML	AIML	2	2	0	0	03	50	50	100	3
3	PEC	21AI73X	Professional Elective Course-II	AIML	AIML	3	0	0	0	03	50	50	100	3
4	PEC	21AI74X	Professional Elective Course-III	AIML	AIML	2	0	0	0	03	50	50	100	2
5	OEC	21AI75X	Open Elective Course-II	AIML	AIML	3	0	0	0	03	50	50	100	3
6	Project	21AIP76	Project Work	-	-	Two contact hours /week				03	100	100	200	10
Total											350	350	700	24

PROFESSIONAL ELECTIVE COURSE -II

21AI731	Advanced Data Security and Privacy	21AI733	High Performance Computing
21AI732	System Architecture and Design	21AI734	Quantum Computing

PROFESSIONAL ELECTIVE COURSE -III

21AI741	Business Intelligence	21AI743	Social Network Analysis
21AI742	Knowledge Representation	21AI744	Applied AI

OPEN ELECTIVES – II (offered by the Department to other Department students)

21AI751	AI in Practice	21AI753	Introduction to Big Data
21AI752	Software Testing	21AI754	Introduction to Data Visualization



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PROJECT WORK:

The objective of the Project work is

- 1) To encourage independent learning and the innovative attitude of the students.
- 2) To develop interactive attitude, communication skills, organization, time management, and presentation skills.
- 3) To impart flexibility and adaptability.
- 4) To inspire team working.
- 5) To expand intellectual capacity, credibility, judgment and intuition.
- 6) To adhere to punctuality, setting and meeting deadlines.
- 7) To instill responsibilities to oneself and others.
- 8) To train students to present the topic of project work in a seminar without any fear, face the audience confidently, enhance communication skills, involve in group discussion to present and exchange ideas.

CIE procedure for Project Work:

Single discipline: The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two senior faculty members of the Department, one of whom shall be the Guide.

The CIE marks awarded for the project work, shall be based on the evaluation of project work Report, project presentation skill, and question and answer session in the ratio 50:25:25. The marks awarded for the project report shall be the same for all the batch mates.

Interdisciplinary: Continuous Internal Evaluation shall be group-wise at the college level with the participation of all guides of the college. Participation of external guide/s, if any, is desirable. The CIE marks awarded for the project work, shall be based on the evaluation of project work Report, project presentation skill, and question and answer session in the ratio 50:25:25. The marks awarded for the project report shall be the same for all the batch mates.

SEE procedure for Project Work: SEE for project work will be conducted by the two examiners appointed by the University. The SEE marks awarded for the project work, shall be based on the evaluation of project work Report, project presentation skill, and question and answer session in the ratio 50:25:25.



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

VIII SEMESTER

	Course and Course Code		Course Title	Teaching Department	Syllabus and QP Setting	Teaching Hours /Week				Examination				Credits
						L	T	P	S	Duration	CIE	SEE	Total	
1	Seminar	21AI81	Technical Seminar	AIML	AIML	One contact hour /week				-	100	-	100	1
2	INT	21INT82	Research Internship/ Industry Internship	-	-	Two contact hour /week				03	100	100	200	15
3	NMC	21NS83	National Service Scheme (NSS)	NSS	NSS	Completed during the intervening period of III Semester to VI semester.				-	-	-	-	0
		21PE83	Physical Education (PE) (Sports And Athletics)	PE	PE									
		21YO83	Yoga	Yoga	Yoga									
4	NMC	21ATP1/ 21ATP2	Activity Points	-	-	0	0	0	0	-	-	-	-	0
Total											200	100	300	16

TECHNICAL SEMINAR:

The objective of the seminar is to inculcate self-learning, present the seminar topic confidently, enhance communication skill, involve in group discussion for exchange of ideas. Each student, under the guidance of a Faculty, shall choose, preferably, a recent topic of his/her interest relevant to the programme of Specialization.

- 1) Carry out literature survey, systematically organize the content.
- 2) Prepare the report with own sentences, avoiding a cut and paste act.
- 3) Type the matter to acquaint with the use of Micro-soft equation and drawing tools or any such facilities.
- 4) Present the seminar topic orally and/or through PowerPoint slides.
- 5) Answer the queries and involve in debate/discussion.
- 6) Submit a typed report with a list of references.



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The participants shall take part in the discussion to foster a friendly and stimulating environment in which the students are motivated to reach high standards and become self-confident.

Evaluation Procedure:

The CIE marks for the seminar shall be awarded (based on the relevance of the topic, presentation skill, participation in the question-and-answer session, and quality of report) by the committee constituted for the purpose by the Head of the Department. The committee shall consist of three teachers from the department with the senior-most acting as the Chairman.

Marks distribution for CIE of the course:

Seminar Report:50 marks Presentation skill:25 marks, Question and Answer: 25 marks.

No SEE component for Technical Seminar

National Service Scheme/Physical Education (Sport and Athletics)/ Yoga:

- 1) In case, any student fails to register for NSS, PE or Yoga and fails to submit the requirements, he/she shall be deemed to have not completed the requirements of the course. In such a case, the student has to fulfill the course requirements during subsequent semesters.
- 2) Successful completion of the course shall be indicated as satisfactory in the VIII semester grade card. Non-completion of the course shall be indicated as Unsatisfactory.
- 3) These courses shall not be considered for vertical progression as well as for the calculation of SGPA and CGPA, but completion of the courses shall be mandatory for the award of degree.

AICTE ACTIVITY POINTS

Sl. No.	Student Category	Activity Points to be Earned	Applicable course code
1	Day college Regular student admitted to the 4 years degree programme	100	19ATP1
2	Student entering 4 years degree programme through lateral entry	75	19ATP2
3	The student transferred from other Universities to the fifth semester	50	19ATP2

AICTE Activity Points to be earned by students admitted to BE/B.Tech./B.Plan day college programme (For more details refer to Chapter 6, AICTE Activity Point Programme, Model Internship Guidelines):

Over and above the academic grades, everyday college regular student admitted to the 4 years Degree programme and every student entering 4 years Degree programme through lateral entry, shall earn 100 and 75 Activity Points respectively for the award of degree



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through AICTE Activity Point Programme. Students transferred from other Universities to the fifth semester are required to earn 50 Activity Points from the year of entry to VTU. The Activity Points earned shall be reflected on the student's eighth semester Grade Card. The activities can be spread over the years, anytime during the semester weekends and holidays, as per the liking and convenience of the student from the year of entry to the programme. However, the minimum hours' requirement should be fulfilled. Activity Points (non-credit) do not affect SGPA/CGPA and shall not be considered for vertical progression.

In case students fail to earn the prescribed activity Points, an Eighth semester Grade Card shall be issued only after earning the required activity Points. Students shall be admitted for the award of the degree only after the release of the Eighth semester Grade Card.



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Assessment Evaluation pattern for standalone theory and Lab courses					
CIE in Theory Course (50 Marks)					
(Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating and Creating)					
Each Test will be conducted for 50 Marks adding up to 150 Marks. Final test marks will be reduced to 30 Marks.	IAT	Marks			
		Max. Marks	Reduced to 30 Marks	Average	IAT Final Marks
	IAT-I	50	30(A)	(A+B+C)/3 =30 (D)	Total out of 30 marks
	IAT-II	50	30(B)		
	IAT-III	50	30(C)		
QUIZ	Evaluated for 30 Marks			Reduced to 10 Marks	
	30			10 (E)	
Alternate Assessment Tool (AAT)	Reflection Note on Guest Lecture/ Reflection note on Industrial Visit/ E-course certification/Building models/Group discussion/case study/Seminar/Paper Presentation/Open Book Assignment				
	10 Marks (F)				
Total CIE Marks	CIE (D) +QUIZ (E)+ AAT(F)			50 Marks	
CIE in Laboratory (50 Marks)					
Conduction of Experiments(E)		30 (G)	Total= G+H=50	Total out of 50 Marks	
Performance of the Experiment (On completion of every experiment/program in the laboratory, the students shall be evaluated and marks shall be awarded on the same day. 20 marks are for conducting the experiment and calculations/observations/output)	20				
Record	05				
Evaluation of Outcome/ Viva	05				



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Final test/Case Study/Open Ended Experiment(if it is not test then a five page report stapled has to be submitted)	50	Reduced to 20 (H)		
Final theory CIE marks =50 Marks Final Lab CIE marks =50 Marks				
* SEE Theory will be conducted for 100 marks *SEE Lab will be conducted for 50 marks				

Assessment Evaluation pattern for AEC/UHV courses					
CIE in Theory Course (50 Marks)					
(Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating and Creating)					
Each Test will be conducted for 50 Marks adding up to 100 Marks. Final test marks will be reduced to 30 Marks.	IAT	Marks			
		Max. Marks	Reduced to 30 Marks	Average	IAT Final Marks
	IAT-I	50	30(A)	(A+B)/2 =30 (C)	Total out of 30 marks
	IAT-II	50	30(B)		
QUIZ	Evaluated for 30 Marks			Reduced to 10 Marks	
	30			10 (D)	
Alternate Assessment Tool (AAT)	Reflection Note on Guest Lecture/ Reflection note on Industrial Visit/ E-course certification/Building models/Group discussion/case study/Seminar/Paper Presentation/Open Book Assignment				
	10 Marks (E)				
Total CIE Marks	CIE (C) +QUIZ (D)+ AAT(E)			50 Marks	
CIE in AEC Laboratory (50 Marks)					
Conduction of Experiments(E)		30 (G)	Total= G+H=50		Total out of



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Performance of the Experiment (On completion of every experiment/program in the laboratory, the students shall be evaluated and marks shall be awarded on the same day. 20 marks are for conducting the experiment and calculations/observations/output)	20			50 Marks
Record	05			
Evaluation of Outcome/ Viva	05			
Final test/Case Study/Open Ended Experiment(if it is not test then a five page report stapled has to be submitted)	50	Reduced to 20 (H)		
Final theory CIE marks =50 Marks Final Lab CIE marks =50 Marks				
* SEE Theory will be conducted for 50 marks *SEE Lab will be conducted for 50 marks				



FOUNDATION IN MATHEMATICS FOR COMPUTING

Course Code	21MAT31	Credits:	03
L:T:P:S	3:0:0:0	CIE Marks: SEE Marks	50:50
Course Type	BSC	Exam Hours	03

Course Objectives

1	To introduce Vector Spaces and Linear Transformation.
2	To introduce to the concept of Statistical Analysis.
3	To provide the foundations of probabilistic, statistical analysis and testing of hypothesis mostly used in varied applications in engineering and science.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand the concept of Vector Spaces and its applications, Eigenvalues and Eigen vectors, the basic knowledge on data collection and various statistical elementary tools, Probability distributions and Testing of Hypothesis.	L1- L4
CO2	Analyze the data using Correlation and Regression, dimension of a vector space, the long-term behaviour of dynamical systems using eigenvalue decompositions, Statistical methods and Probability theory and the relevant real-life problems from testing of hypothesis.	L1- L4
CO3	Apply the basic properties of subspaces and vector spaces, diagonalization to compute matrix powers, theoretical discrete and Continuous probability distributions, Statistical methods and different testing tools in Engineering problems.	L1- L4
CO4	Evaluate the dependence relations between linearly dependent vectors, eigenspace for a matrix, random variables and probability distributions, Statistical modelling in computer applications, problems that are faced in testing of a hypothesis with reference to the errors in decision making in Engineering field.	L1- L4
CO5	Solve Differential equation and Discrete Dynamical system using the concept of Eigenvalues and Eigenvectors, provide an axiomatic description of an abstract vector space, Statistical Modelling, Probability and Testing of Hypothesis in the relevant application problems.	L1- L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	3	-

Module	Topic	Hours	Assesment	CO
1	Vector Spaces: Vector spaces and Subspaces, Null Spaces, Column Spaces and Linear Transformation, Linearly independent sets, Bases, Dimension of VectorSpace, Rank- Case Study.	8	CIE+SEE+QUIZ	CO1- CO5



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	Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111102011 Video Lectures:			
2	Eigenvalues and Eigenvectors: Review of Characteristic Equation, Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Transformation, Quadratic to Canonical-Case Study. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111107112 Video Lectures:		CIE+SEE+QUIZ	CO1-CO5
3	Statistics: Analyzing a data – Mean, Mode, Median, Standard Deviation-Combination of two groups, correlation, Covariance, Linear regression- Case Study. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111104120 Video Lectures:		CIE+SEE+QUIZ	CO1-CO5
4	Probability Distributions: Types of variables, Probability Mass & Density function, Mean & Variance of random and continuous variable, Geometric distribution, Poisson distribution & Exponential and Normal distributions- Case Study. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111104032 Video Lectures:		CIE+SEE+QUIZ	CO1-CO5
5	Sampling Distribution and Optimization Techniques: Sampling distribution, Central limit theorem, Sampling with & without replacement, Confidence limits for means, Student's t- distribution. Convex Optimization, Gradient Descent algorithms - Batch, Mini batch, Stochastic, their performance comparison- Case Study. Pedagogy: Chalk and talk/PowerPoint Presentation/Video Web Link: https://nptel.ac.in/courses/111104073 Video Lectures:		CIE+SEE+QUIZ	CO1-CO5
	SCILAB: 1. Program on Vector Spaces and Subspaces. 2. Program on Linear Independence and Basis for a Vector Space. 3. Program on Eigen Values and Eigen Vectors. 4. Program on Diagonalization. 5. Program on Correlation.			CO1-CO5

Text Books

1	B.S. Grewal, "Higher Engineering Mathematics" Khanna Publishers, 43 rd Edition, 2014June, ISBN: 9788174091956.
2	Erwin Kreyszig; Advanced Engineering Mathematics; John Wiley & Sons, 9th Edition, 2007, ISBN: 9788126531356.
3	Gilbert Strang, Linear Algebra and its Applications, 4 th edition, Cengage Publishers, 2014, ISBN: 9788131501726. ISBN: 9788177583335.

Reference Books

1	B.V.Ramana, "Higher Engineering Mathematics", Tata McGraw-Hill, 2006; ISBN: 9780070634190.
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2	Kenneth Hoffmann, Ray Kunze, Linear Algebra, 2Nd Edition, Prentice Hall India Learning Private Limited, 1978. ISBN-13: 978-0135367971.
3	Schaum's Outline: Introduction to Probability and Statistics, McGraw Hill Education (India) Private Limited (1 September 2005); ISBN-13: 978-0070605015.
4	S.D.Sharma, Operations Research, Kedarnath Ramnath & Co 2002, ISBN : 978-93- 85935-14-5.



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DATA STRUCTURES WITH APPLICATIONS

Course Code	21AI32	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives

1	Explain fundamentals of data structures and their applications essential for programming / problem solving
2	Analyze Linear Data Structures: Stack, Queues, and Lists.
3	Analyze Non-Linear Data Structures: Trees, Graphs.
4	Assess appropriate data structure during program development / problem solving.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Apply the concept of data structures to solve problems.	L3
CO2	Develop applications using stacks and queues.	L3
CO3	Analyze and implement operations on linked lists and demonstrate their applications.	L4
CO4	Demonstrate the implementation of various types of Binary Search Trees and their traversals order.	L4
CO5	Perform various operations on tree, AVL tree & RB tree.	L3
CO6	Illustrate tree traversal methods to manage engineering data effectively.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO4	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-

Module	Topic	Hours	Assessment	CO
1	Basic Concepts: Introduction to Data Structure and its classification, the need for Data structure, Algorithm specification, performance analysis & measurements, Pointer, Array, Structure and union, Sparse Matrices, dynamic memory allocation.	10	CIE+SEE+QUIZ	CO1
2	Stacks And Queues: Introduction to Stacks, Stacks Using Dynamic Arrays, Conversion of Infix expression to Postfix expression, Evaluation of Postfix Expressions, Introduction to Queues, Types of Queue: Ordinary queue, Circular Queues & Double ended queue, Application of stacks and Queues.	10	CIE+SEE+QUIZ	CO2
3	Linked Lists: Definition of Linked lists and Chains, Types of Linked List: Singly Linked List, Circular Singly Linked List, Doubly Linked Lists & Circular doubly linked list, Application of Linked List.	10	CIE+SEE+QUIZ	CO3



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4	Trees: Introduction to Binary Trees, Properties of Binary Tree, Operation on Binary Tree, Traversals in Binary Trees, Binary Search Tree, Expression Trees, Heaps.	10	CIE+SEE+QUIZ	CO4
5	Efficient Binary Search Trees: Optimal Binary Search Trees, AVL Trees, Construction of AVL tree, Red-Black Trees, Construction of red black tree Graphs: Representation of Graphs - Adjacency/ Cost Matrix, Adjacency Lists, and Traversal of Graphs (BFS and DFS)	10	CIE+SEE+QUIZ	CO5

Text Books

- | | |
|---|---|
| 1 | Horowitz, Sahni, Anderson-Freed: Fundamentals of Data Structures in C, 2nd Edition, Universities Press, 2007. |
|---|---|

Reference Books

- | | |
|---|---|
| 1 | Yedidyah, Augenstein, Tannenbaum: Data Structures Using C and C++, 2nd Edition, Pearson Education, 2003. |
| 2 | Richard F. Gilberg and Behrouz A. Forouzan: Data Structures A Pseudocode Approach with C, Cengage Learning, 2005. |
| 3 | A.M Padma Reddy," Approach of Data Structures", Person Publication, 5th Edition, 2015. |



DATABASE MANAGEMENT SYSTEM

Course Code	21AI33	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives

1	Understand the basic concepts and the applications of database systems.
2	Master the basics of SQL and construct queries using SQL
3	Understand the relational database design principles.
4	To understand the concepts of normalization.
5	Design and build simple database applications.
6	Familiar with the basic issues of transaction processing and concurrency control.

Course Outcomes	Blooms Taxonomy(highest level)
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CO1	Understand the basics of databases and database management systems	L2
CO2	Apply DBMS concepts to design and create databases that address specific real-world scenarios	L3
CO3	Analyze a given scenario and use suitable database technique	L4
CO4	Demonstrate the various SQL & Relational algebra query processing	L4
CO5	Ability to conduct experiments as individual or team to using modern tools like MySQL and Mongo DB	L4
CO6	Write clear and concise experiment reports that detail the methods, results, and conclusions of DBMS experiment	L3

CO-PO-PSO Mapping	
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[illegible]

Module	Topic	Hours	Assessment	CO
1	Introduction to Databases: Introduction, An Example, Characteristics of Database approach, Advantages of using DBMS approach, When not to use a DBMS Database System Concepts and Architecture: Data models, Schemas and instances, Three schema architecture and data independence Database languages and interfaces, The database system environment. SQL: SQL Data Definition and Data Types specifying basic constraints in SQL, Basic retrieval queries in SQL, Insert, Delete and Update statements in SQL, Additional features of SQL, More complex	8	CIE+SEE+QUIZ	CO1

	SQL Queries, Specifying Constraints as Assertions and Triggers, Views (Virtual Tables) in SQL, Schema Change Statement in SQL.			
2	Data Modelling using the Entity-Relationship(ER) model: Using High-Level Conceptual Data Models for Database Design, A sample Database Application, Entity types, Entity Sets, Attributes, and Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity types, Refining the ER Design, ER Diagrams, Naming Conventions and Design Issues, Relationship Types of Degree Higher than two, Relational Database Design using ER-to-Relational Mapping	8	CIE+SEE+QUIZ	CO2-CO4
3	Relational Data Model and Relational Database Constraints: Relational Model Concepts, Relational Model Constraints and Relational Database Schemas, Update Operations, Transactions and Dealing with Constraint Violations. Relational Algebra: Unary Relational Operations, SELECT, and PROJECT, Relational Algebra Operations from Set Theory Binary Relational Operations: JOIN and DIVISION, Additional Relational Operations, Examples of Queries in Relational Algebra.	8	CIE+SEE+QUIZ	CO3-CO4
4	Database Design Theory and Normalization: Informal Design Guidelines for Relation Schemas, Functional Dependencies, And Normal Forms Based on Primary Keys, General Definitions of Second and Third Normal Forms, Boyce-Codd Normal Form, Fourth Normal Form, Fifth Normal Form.	8	CIE+SEE+QUIZ	CO2-CO5
5	Transaction Processing, Concurrency Control, and Recovery: Introduction to Transaction Processing, Transaction and System Concepts, Desirable Properties of Transactions, Two-Phase Locking Techniques for Concurrency Control, Recovery Concepts, NO-UNDO/REDO Recovery Techniques based on Deferred Update, Recovery Techniques Based on Immediate Update, Shadow Paging, The ARIES Recovery Algorithm. Mongo DB: CRUD & nesting, Indexing, Aggregation, Map reduce, Replica set, Sharding, Geospatial and GridFS	8	CIE+SEE+QUIZ	CO4

Text Books	
1	“Fundamental of Database Systems”, Ramez Elmasri and Shamkant B Navathe ,Sixth Edition, Pearson, 2019
2	“Database Management Systems”, Ramakrishnan and Gehrke , Third Edition, McGraw Hill, 2016
3	“Seven Databases in Seven Weeks “,Luc Perkins with Eric Redmond and Jim R. Wilson, Second edition, 2018

Reference Books	
1	“An Introduction to Database Systems”, C.J.Date, A.Kannan, S.Swamynathan,8th Edition ,Pearson Education,2017
2	“Database Systems: The Complete Book “, Hector Garcia Molina, Jeffrey D. Ullman, Jennifer Widom ,Second Edition, Pearson Education, 2018
3	“Database System Concepts”. Abraham Silberschatz, Henry F. Korth, S. Sudarsha , Sixth Edition ,Tata McGraw-Hill, 2017.



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SOFTWARE ENGINEERING

Course Code	21AI34	Credits:	03
L:T:P:S	2:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	03

Course Objectives	
1	Outline software engineering principles and activities involved in building large software programs.
2	Introduce Software design using UML tools and various system models.
3	Introduce Agile Software development and Agile methods.
4	Introduce Software Testing and Software Evolution processes in detail.
5	Explain project planning, Cost estimation techniques and Software quality, standards and metrics.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Illustrate the fundamentals of Software Engineering, Software process models, and Requirements Engineering.	L2
CO2	Apply Agile software development and Agile methods.	L3
CO3	Apply and Analyse Software Testing methods.	L3
CO4	Discuss the Project Planning process, Software Quality standards, and business strategies.	L3

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Introduction: Professional Software Development - Software Engineering, Case studies. Software Processes: Software process models - Waterfall Model, Incremental Model and Spiral Model, Software Process Activities Requirement Engineering: Requirements Specification, Requirement elicitation and analysis, Requirements Validation, Requirements Management, Functional Specification and Non-Functional Requirements.	8	CIE+SEE+QUIZ	CO1
2	Software Design and Modeling Concepts: What is OO orientation? What is OO development, OO themes, Modeling as design technique, modeling, abstraction, The three models, object and class concepts , Links and association concepts, A sample class model. System Models: Context models. Interaction models. Structural models. Behavioral models. Model-driven engineering	8	CIE+SEE+QUIZ	CO2

3	Agile Software Development: Agile methods, Plan-driven and agile development, Extreme programming, Agile project management, Scaling agile methods. SCRUM Methodology, SCRUM. Project Scheduling: Basic Concepts, Project Scheduling, Defining a Task set for the software Project, Defining a Task network, Scheduling.	8	CIE+SEE+QUIZ	CO2,3
4	Software Testing: Development testing, Test-driven development, Release testing, User testing, Dataflow testing and Path Testing, Case Study: Commission Problem Software Evolution: Evolution processes, Program evolution dynamics, Software maintenance, Legacy system management	8	CIE+SEE+QUIZ	CO1,4
5	Project planning: Software pricing, Plan-driven development, Project scheduling, Agile planning, Estimation techniques, COCOMO cost modeling. Case Study on Project Planning and Management Quality management: Software quality, Software standards, Reviews and inspections, Software measurement and metrics Business Strategy: The concept of strategy, Introduction and objectives, Role of Strategy in success, Framework of strategy analysis, Brief history of Business strategy, The Strategy process.	8	CIE+SEE+QUIZ	CO4

Text Books	
1	“Software Engineering”, International Computer Science Series, Ian Somerville, Pearson Education, 10th Edition, 2020.
2	Michael Blaha, James Rumbaugh, Object Oriented Modelling and Design with UML, , Pearson Education, 2nd Edition, 2017.
3	“Software Testing – Principles and Practices”, Srinivasan Desikan and Gopalaswamy Ramesh, Pearson, 2018.

Reference Books	
1	“Software Engineering: Principles and Practice”, Hans van Vliet, Wiley India, 3rd Edition, 2018.
2	“Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling” by By Jennifer Davis, Ryn Daniels, O’Reilly Publications, 2018
3	“Software Engineering: A Practitioner’s Approach”, Roger S Pressman, McGraw Hill, 6th Edition 2015.
4	The SCRUM Primer, Ver2.0, http://www.goodagile.com/scruprimer/scruprimer20.pdf .



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APPLICATION DEVELOPMENT USING PYTHON

Course Code	21AI35	Credits:	01
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCCL	Exam Hours	03

Course Objectives	
1	Apply various data preprocessing techniques using Python for cleaning, transforming, and organizing datasets for machine learning applications.
2	Integrate their machine learning models with popular Python libraries such as NumPy, Pandas, and scikit-learn.
3	Perform exploratory data analysis and advanced data visualization techniques using Python libraries such as Matplotlib and Seaborn.
4	Design and develop small applications using advance python libraries.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply Python programming skills to preprocess and clean real-world datasets for machine learning tasks	L3
CO2	Ability to conduct experiments using modern tools like Tensorflow and Keras	L3
CO3	Ability to analyze outputs of machine learning models and Perform statistical hypothesis testing to validate the significance of machine learning results	L4
CO4	Present experimental findings in writing, to convey the importance and relevance of the experiments.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-

Module	Topic	Hours	Assessment	CO
1	Design and Implement a Python program to accept 3 digits from the user and print all possible combination from digits. Create a Python program to take two command line inputs and compute the GCD	2	CIE+SEE+QUIZ	CO1 ,2,3
2	A list rotation consists of taking the last element and moving it to the front. For instance, if we rotate the list [1,2,3,4,5], we get [5,1,2,3,4]. If we rotate it again, we get [4,5,1,2,3]. Write a Python function rotatelist(ls,k) that takes a list ls and a positive integer k and returns the list ls after k rotations. If k is not positive, your function should return ls unchanged. Note that your function should not change ls itself, and should return the rotated list. Here are some examples to show how your function should work. >>>rotatelist([1,2,3,4,5],1)	2	CIE+SEE+QUIZ	CO1 ,2,3



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	#output is [5, 1, 2, 3, 4] >>>rotatelist([1,2,3,4,5],3) #output is [3, 4, 5, 1, 2] >>>rotatelist([1,2,3,4,5],12) #output is [4, 5, 1, 2, 3] Design and implement a python code that accepts two string from user and displays the characters which are present in both the strings. Use Set sequence type to achieve the same			
3	Design a python program to compute the number of characters, words and lines in a file. Also Print the most frequent words read from the file. Apply import, from, * and other module related concepts to create a module called “calc” consists of 4 function that should return sum, division, multiplication and subtraction. Create another module caller “user”, import the calc module and illustrate the use of all the functions of calc module.	2	CIE+SEE+QUIZ	CO1 ,2,3
4	Design & implementation of programs in python to Manipulate image.	2	CIE+SEE+QUIZ	CO1 ,2,3
5	Data Science and Python: Discovering the match between data science and python: Understanding the role of programming in Data Science, Creating the Data Science Pipeline: Preparing the data, Performing exploratory data analysis, Learning from data, Visualizing, Obtaining insights and data products .Introducing Python's Capabilities and Wonders: Why Python?, Flavors of Python, Python as a calculator, Working with Python, Getting a taste of the language, Understanding the need for indentation, Working at the command line or in the IDE:IDLE,Notepad,	2	CIE+SEE+QUIZ	CO1 ,2,3

Text Books

1	B.S. Grewal, “Higher Engineering Mathematics” Khanna Publishers, 43 rd Edition, 2014June, ISBN: 9788174091956.
2	Erwin Kreyszig; Advanced Engineering Mathematics; John Wiley & Sons, 9th Edition, 2007, ISBN: 9788126531356.
3	Gilbert Strang, Linear Algebra and its Applications, 4thedition, Cengage Publishers, 2014, ISBN: 9788131501726. ISBN: 9788177583335.

Reference Books

1	B.V.Ramana, “Higher Engineering Mathematics”, Tata McGraw-Hill, 2006; ISBN: 9780070634190.
2	Kenneth Hoffmann, Ray Kunze, Linear Algebra, 2Nd Edition, Prentice Hall India LearningPrivate Limited, 1978. ISBN-13: 978-0135367971.
3	Schaum's Outline: Introduction to Probability and Statistics, McGraw Hill Education (India) Private Limited (1 September 2005); ISBN-13: 978-0070605015.
4	S.D.Sharma, Operations Research,Kedarnath Ramnath & Co 2002, ISBN : 978-93- 85935-14-5.



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SOCIAL CONNECT & RESPONSIBILITY

Course Code	21SCR36	Credits:	01
L:T:P:S	1:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	SCR	Exam Hours	03

Course Objectives

1	Enable the student to do a deep drive into societal challenges being addressed by NGO(s), social enterprises & the government.
2	Build solutions to alleviate the complex social problems through immersion, design & technology.
3	Provide a formal platform for students to communicate and connect with their surroundings.
4	Enable to create of a responsible connection with society.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand social responsibility	L2
CO2	Practice sustainability and creativity	L2
CO3	Showcase planning and organizational skills	L2

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
1	Plantation and adoption of a tree: Plantation of a tree that will be adopted for four years by a group of Students. They will also make an excerpt either as a documentary or a photo blog describing the plant's origin, its usage in daily life, and its appearance in folklore and literature. Field visits, Case studies and practice sessions	3	CIE+SEE+QUIZ	CO1 ,2,3
2	Heritage walk and crafts corner: Heritage tour, knowing the history and culture of the city, connecting to people around through their history, knowing the city and its craftsman, photo blog and documentary on evolution and practice of various craft forms. Field visits, Case studies and practice session	3	CIE+SEE+QUIZ	CO1 ,2,3
3	Organic farming and waste management: usefulness of organic farming, wet waste management in neighboring villages, and implementation in the campus. Field visits, Case studies and practice session	3	CIE+SEE+QUIZ	CO1 ,2,3
4	Water Conservation: Knowing the present practices in surrounding villages and implementation in the campus, documentary or photo blogs presenting the current practices. Field visits, Case studies and practice sessions.	3	CIE+SEE+QUIZ	CO1 ,2,3



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5	Food Walk City's culinary practices, food lore, and indigenous materials of the region used in Cooking. Field visits, Case studies and practice sessions	3	CIE+SEE+QUIZ	CO1 ,2,3
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Text Books	
1	A foundation course in HUMAN VALUES and Professional ethics; presenting a universal approach to value education- through self-exploration by, R R Gaur, R Sangal & GP Bagaria, Excel books Pvt. Ltd.
2	Professional Ethics & Human Values: Prof. D.R. Kiran, TATA Mc Graw Hill Education

Reference Books	
1	Human Values: A. N. Tripathy (2003, New Age International Publishers)
2	Ethics in Engineering Mike W. Martin, Department of Philosophy, Chapman University and Roland Schinzinger, School of Engineering, University of California, Irvine.



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ROBOTIC PROCESS & AUTOMATION LAB

Course Code	21AIL381	Credits:	01
L:T:P:S	0:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	03

Course Objectives

1	Understand the Future of Robotic Process Automation (RPA).
2	Design and implement workflows for automating simple tasks.
3	Build End-to-End Processes with Automation

Course Outcomes

Course Outcomes		Blooms Taxonomy
CO1	Demonstrate the knowledge of the bright and exciting future of Robotic Process	L2
CO2	Analyze business process automations and identify potentials use cases	L4
CO3	Create workflows with good understanding of Recorders, Editor, and various essential Commands to build simple tasks / Bots for automating simple process	L5
CO4	Using AA tool – develop and demonstrate end-to-end process with industry standard coding techniques.	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
1	Introduction to RPA- What is Robotic Process Automation? Scope and Techniques of automation, Understanding Enterprise Processes, Robotic Process Automation (RPA) - What can RPA DO? Benefits of RPA, Components of RPA, RPA Platform. Applications of RPA: - Differentiating the RPA from Automation Programming Review, Programming fundamentals from an RPA perspective, Future of automation.	8	CIE+SEE+QUIZ	CO1 ,2,3
2	Automation Anywhere: Introduction to AA, Automation Anywhere Enterprise Tool – An introduction AA Control Room, Control Room Settings, License Settings, All Menus, Theory, demonstration and hands on practice and experience on the system Recorders-Web Recorder, Screen Recorder, Smart Recorder, AA Commands- Read from CSV/Text, Excel, Database, Files/Folder, Error Handling, String Operation, Variables, Variable Operation, PDF Integration, Email Automation, OCR, Web Recorder	8	CIE+SEE+QUIZ	CO1 ,2,3

Text Books

1	Automation Anywhere Enterprise, Exercise Handbook V1.0 2. https://resources.automationanywhere.com/ebook
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DISCRETE MATHEMATICAL STRUCTURES

Course Code	21MAT41A	Credits: 03	
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	BSC	Exam Hours	03

Course Objectives

1	To introduce the basics of Set Theory, Number Theory and some of its applications.
2	To understand and apply problem-solving skills by using truth tables, basic logical connectives and proofs.
3	To understand the Properties of Relations and various types of functions also graph theory concepts to apply some of its applications in computer science.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand basic concepts of Set Theory & Number Theory, Fundamental logics, functions, Relations and Graph Theory.	L2
CO2	Analyze the logic for validity of the given arguments, sets and perform operations and algebra on sets, properties of Functions, Relations and Graph Theory.	L4
CO3	Apply the concepts of Relations, Hasse Diagram and Poset, Mathematical Logics, Functions and Number Theory, Graph theory in Engineering Problems.	L3
CO4	Evaluate functions like Stirling numbers using fundamental principles, logical propositions via truth tables, combinations and permutations on sets, relations and digraphs, trees in Engineering field.	L3
CO5	Solve the engineering problems using mathematical logic, recurrence relations, even and odd integers, rational and irrational numbers and divisibility using Number Theory, connection between bijective functions and real life problems using graphs and trees.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
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1	Set Theory and Number Theory: Sets and Subsets, Set Operations and the Laws of Set Theory, Addition Principles, Concept of Number Theory-Simple ProblemsCase study. Application: Representation of Set Theory problems using Venn diagrams Pedagogy: Chalk and talk/PowerPoint Presentation/Videos	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4
	Web Link: https://onlinecourses.nptel.ac.in/noc20_ma42/preview Video Lectures :			
2	Fundamentals of Logic: Basic Connectives and Truth Tables, Logical Equivalence – The Laws of Logic, Logical Implication– Rules of Inference, Quantifiers with one variable- Case study. Application: AND- OR gate problems for real time. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/106102013 Video Lectures :	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4
3	Relations: Properties of Relations, Computer Recognition: ZeroOne Matrices and Directed Graphs, Partial Orders – Hasse Diagrams, Equivalence Relations and Composition of RelationsCase study. Application: Recurrence relation to solve Tower of Hanoi, Fibonacci numbers and Rabbit Population Problem. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://archive.nptel.ac.in/courses/111/107/111107058/ Video Lectures:	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4
4	Functions: Cartesian Products, Functions – Plain and One-to- One, Onto Functions – Stirling Numbers of the Second Kind, Function Composition- Case study. Application: Pigeon Hole Principle for allocations. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111107058 Video Lectures :	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4
5	Graph Theory & Its Applications: Definitions and Examples, Sub graphs, Complements, and Graph Isomorphism, Euler Trails and Circuits, Hamiltonian paths, cycles, Basics of Graph Coloring-Case study. Application: Coloring of Maps, Solving Sudoko, Coloring of graphs resulting in Bipartite graphs. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://onlinecourses.nptel.ac.in/noc20_ma05/preview Video Lectures:	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4
	Student development program: Workshop on Lattice theory. Self-Study component: Module 1: Cardinality Properties of Set Theory and Applications of Number Theory Module 2: Validity of arguments in propositional calculus and Uses of Logics Module 3: Classification of function and Inverse Function. Module 4: Relational matrix and graph of a relation Module 5: Graph Homomorphism and its applications.			

Text Books

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|---|--|
| 1 | Ralph P. Grimaldi: Discrete and Combinatorial Mathematics, 5th Edition, Pearson Education. 2006, ISBN-10. 9788177584240. |
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2	B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 43rd Edition, 2014 June, ISBN: 9788174091956.
3	David Burton, Elementary Number Theory 7th Edition, Mc Graw Hill Higher Education 2006 ISBN 978-0-07-338314-9.

Reference Books	
1	Ralph P. Grimaldi: Discrete and Combinatorial Mathematics, 5th Edition, Pearson Education. 2006, ISBN-10. 9788177584240.
2	B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 43rd Edition, 2014 June, ISBN: 9788174091956.
3	David Burton, Elementary Number Theory 7th Edition, Mc Graw Hill Higher Education 2006 ISBN 978-0-07-338314-9.



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DESIGN & ANALYSIS OF ALGORITHMS

Course Code	21AI42	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives	
1	To learn the methods for analyzing algorithms and evaluating their performance.
2	To demonstrate the efficiency of algorithms using asymptotic notations.
3	To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound.
4	To learn the concepts of P and NP complexity classes.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply Brute Force Techniques to solve the Numerical, String and Graph problems.	L3
CO2	Apply Recursive Algorithms towards optimal solution for real time problems.	L3
CO3	Analyze the time complexity of the P Class problems.	L4
CO4	Analyze the time complexity of the NP and NP Hard Class problems	L4
CO5	Implement and analyze the efficiency of algorithmic design techniques in solving Real time problems using open source tools.	L4
CO6	Write clear and concise experiment reports that detail the methods, results, and Analysis of algorithm techniques.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3		-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	3	-	-	-	-	-	-	-	-	-	-	3	-	-
CO6	-	3	-	-	-	-	-	-	2	2	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Introduction: What is an Algorithm?, Algorithm Specification , Analysis Framework), Performance Analysis: Space complexity, Time complexity . Asymptotic Notations: Big-Oh notation (O), Omega notation (Ω), Theta notation (Θ), and Little-oh notation (o), Mathematical analysis of NonRecursive and recursive Algorithms with Examples . Brute Force Approaches: Selection Sort and Bubble Sort, Sequential Search and Brute Force String Matching, Exhaustive Search (Travelling Salesman problem and Knapsack Problem).	8	CIE+SEE+QUIZ	CO1-CO4



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2	Divide and Conquer: General method, Binary search, Recurrence equation for divide and conquer, Finding the maximum and minimum , Merge sort, Quick sort , Strassen's matrix multiplication , Advantages and Disadvantages of divide and conquer. Decrease and Conquer Approach: Insertion Sort, , Topological Sort	8	CIE+SEE+QUIZ	CO1-CO4
3	Greedy Method: General method, Coin Change Problem, Knapsack Problem, Job sequencing with deadlines. Minimum cost spanning trees: Prim's Algorithm, Kruskal's Algorithm. Single source shortest paths: Dijkstra's Algorithm. Optimal Tree problem: Huffman Trees and Codes Transform and Conquer Approach: Heaps and Heap	8	CIE+SEE+QUIZ	CO1-CO4
4	Dynamic Programming: General method with Examples, Transitive Closure: Warshall's Algorithm, All Pairs Shortest Paths: Floyd's Algorithm, Optimal Binary Search Trees, Knapsack problem, Bellman-Ford Algorithm, Travelling Sales Person problem.	8	CIE+SEE+QUIZ	CO1-CO4
5	Backtracking: General method, N-Queens problem, Sum of subsets problem, Graph coloring, Hamiltonian cycles. Branch and Bound: Assignment Problem, Travelling Sales Person problem, 0/1 Knapsack problem: LC Branch and Bound solution, FIFO Branch and Bound solution. NP Complete and NP-Hard problems: Basic concepts, nondeterministic algorithms, P, NP, NP-Complete, and NP-Hard classes.	8	CIE+SEE+QUIZ	CO1-CO4

Text Books	
1	Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.
2	Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekaran, 2nd Edition, 2014, Universities Press.

Reference Books	
1	Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
2	Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)



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MACHINE LEARNING ESSENTIALS

Course Code	21AI43	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives	
1	Understand the basics of machine learning, including different types of learning and how models work.
2	Practice using important machine learning algorithms like linear regression and decision trees with real-world data.
3	Develop skills to clean and prepare data for analysis, including techniques for handling missing values and creating useful features.
4	Learn how to assess model performance and make improvements to increase accuracy and effectiveness.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand the foundation of Machine learning and statistical methods in Machine Learning	L3
CO2	Apply linear and logistic regression techniques in appropriate problem scenarios	L3
CO3	Use decision trees to solve real world problems	L4
CO4	Analyze the dimensionalities of various classification and clustering techniques	L4
CO5	Apply data analytics and feature engineering techniques on dataset	L4
CO6	Apply regression analysis, classification, clustering, time series analysis and ensemble learning techniques	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	3	-	-	-	-	-	-	-	2	2	-	-	-	2	-

Module	Topic	Hours	Assessment	CO
1	Introduction: Applications of machine learning, Branches of ML, Types of Machine Learning, Machine Learning Process, Data types and Data Analytics for Machine learning, Statistical tests: Mean, Median, Mode, Standard Deviation, 5-point Summary, Student t test, Z test, Chi- square test, Correlation, ANOVA, Hypothesis Testing	8	CIE+SEE+QUIZ	CO1



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2	Linear Regression: Introduction, Advantages and Disadvantages, Applications, Simple Linear Regression, Train Test split, Cost function, Multiple Linear Regression, Model Interpretation, Goodness of fit, Model Selection, Assumptions of Linear Regression, Feature Engineering - Categories as predictors – Dummy Variable, Label Encoding, Interaction, Scaling, Standardisation, Missing value treatment, Outlier Treatment, Feature Extraction, Transformation	8	CIE+SEE+QUIZ	CO2
3	Logistic regression: Why not linear regression? (S Shaped Curve) Need for logistic regression, Introduction, Advantages and Disadvantages, Applications, The Logistic model: Estimating the Regression Coefficients, Making Predictions. Goodness of fit-i) Confusion matrix ii) Precision iii) Recall iv) F1 Score v) ROC / AUC Handling Imbalance class	8	CIE+SEE+QUIZ	CO2
4	Decision Tree: Decision Tree learning Model, Entropy, GINI, Working of single Decision Tress, CART and CHAID algorithm, validating and Pruning of Decision Trees, Probability based Learning, Bayes Theorem, Classification using Bayes Model, Naïve Bayes Algorithm for Continuous Attributes, Other Naïve Bayes Classifiers	8	CIE+SEE+QUIZ	CO3
5	Clustering: Introduction to clustering, Applications of clustering, Distance Measures, Euclidean, Hamming, Manhattan, Minkowski. Hierarchical Clustering, K Means Algorithm, K Mode, K Prototype, DBSCAN (Density based Clustering), Cluster validation Dimensionality Reduction - PCA (Principal Component Analysis)	8	CIE+SEE+QUIZ	CO4

Text Books	
1	Mitchell, T.M. and Tom, M. (1997) Machine Learning. McGraw-Hill, New York.



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OPERATING SYSTEM

Course Code	21AI44	Credits:	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	03

Course Objectives	
1	To understand the importance of operating systems in utilizing the hardware and software resources along with the protection and security mechanisms.
2	To comprehend and analyze various resource management techniques.
3	To implement resource management algorithms.
4	To gain knowledge on the various protection and security mechanisms provided by operating system.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Gain extensive knowledge on principles and modules of Operating Systems	L3
CO2	Apply suitable algorithms for Process Scheduling and Process Synchronization	L3
CO3	Apply suitable techniques for efficient Memory management and Storage management	L3
CO4	Realize the different concepts of OS in platform of usage through case studies.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3		-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	2	-	-

Module	Topic	Hours	Assessment	CO
1	Overview-Introduction: What operating systems do; Operating System Structure, Operating System operations. System Structures: Operating System Services; System calls; Types of System calls.	8	CIE+SEE+QUIZ	CO1
2	Process Management-Process Concept: Process concept; Process scheduling; Process Scheduling: Basic concepts, process creation and termination; scheduling criteria; Scheduling algorithms: FCFS, Pre-emptive and Non-preemptive SJF, Priority Scheduling and Round Robin Scheduling	8	CIE+SEE+QUIZ	CO2



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3	Process Coordination- Synchronization: The Critical section problem, Peterson solution, mutex locks, Semaphores; Classical problems of synchronization: Bounded-Buffer Problem, Reader's-Writer's Problem. Deadlocks: System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock.	8	CIE+SEE+QUIZ	CO2
4	Memory Management-Memory Management Strategies: Background; Contiguous memory allocation; Paging; Virtual Memory management: Background; Demand paging; Page replacement Algorithms: FIFO, LRU, OPTIMAL	8	CIE+SEE+QUIZ	CO3
5	Storage Management-File System: File concept; Access methods; Protection. Secondary Storage Structures: Overview of Mass storage structures; Disk scheduling algorithms. System Protection: Goals of protection, Principles of protection, Domain of protection, Access Matrix.	8	CIE+SEE+QUIZ	CO3

Text Books	
1	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne: Operating System Principles, 8th edition, Wiley India, 2009

Reference Books	
1	D.M. Dhamdhere: Operating systems - A concept based Approach, 2nd Edition, Tata McGraw- Hill, 2002
2	P.C.P. Bhatt: Introduction to Operating Systems: Concepts and Practice, 2nd Edition, PHI, 2008. Harvey M Deital: Operating systems, 3rd Edition, Pearson Education, 1990.
3	Harvey M Deital: Operating systems, 3rd Edition, Pearson Education, 1990



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BIOLOGY FOR ENGINEERS

Course Code	21BE45	Credits:	02
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	03

Course Objectives	
1	To familiarize the students with the basic biological concepts and their engineering applications.
2	To enable the students with an understanding of biodesign principles to create novel devices and structures.
3	To provide the students an appreciation of how biological systems can be re-designed as substitute products for natural systems
4	To motivate the students to develop the interdisciplinary vision of biological engineering.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Elucidate the basic biological concepts via relevant industrial applications and case studies.	L2
CO2	Evaluate the principles of design and development, for exploring novel bioengineering projects.	L3
CO3	Corroborate the concepts of biomimetics for specific requirements.	L3
CO4	Think critically about exploring innovative biobased solutions for socially relevant problems.	L3

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	-	-	1	2	-	-	-	-	2	-	-	-
CO2	2	2	3	-	-	1	2	-	-	-	-	2	-	3	-
CO3	2	3	2	-	-	1	1	1	-	-	-	3	-	3	-
CO4	3	2	1	-	-	1	3	1	-	-	-	2	-	-	-

Module	Topic	Hours	Assessment	CO
1	BIOMOLECULES AND THEIR APPLICATIONS (QUALITATIVE) Introduction to Biology: About biological molecules (Carbohydrates, nucleic acids, proteins, Lipids). Carbohydrates (cellulose-based water filters, PHA and PLA as bioplastics), Nucleic acids (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), Proteins (Proteins as food – whey protein and meat analogs, Plant-based proteins), lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, lignolytic enzyme in bio-bleaching).	8	CIE+SEE+QUIZ	CO1



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2	HUMAN ORGAN SYSTEMS AND BIO DESIGNS - 1 (QUALITATIVE): Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for various diseases). The eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signaling - ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators).	8	CIE+SEE+QUIZ	CO2
3	HUMAN ORGAN SYSTEMS AND BIO-DESIGNS - 2 (QUALITATIVE): Lungs as purification systems (architecture, gas exchange mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems). Muscular and Skeletal Systems as scaffolds (architecture, mechanisms, bioengineering solutions for muscular dystrophy and osteoporosis).	8	CIE+SEE+QUIZ	CO2,3
4	NATURE-BIOINSPIRED MATERIALS AND MECHANISMS (QUALITATIVE): Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircraft), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Sharkskin (Friction reducing swimsuits), Kingfisher beak (Bullet train).	8	CIE+SEE+QUIZ	CO1,4
5	TRENDS IN BIOENGINEERING (QUALITATIVE): Bioprinting techniques and materials, 3D printing of ear, bone, and skin. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging, and Artificial Intelligence for disease diagnosis. Self-healing Bioconcrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, and Arsenic).	8	CIE+SEE+QUIZ	CO4

Text Books
1 Human Physiology, Stuart Fox, Krista Rompolski, McGraw-Hill eBook. 16th Edition, 2022
2 Biology for Engineers, Thyagarajan S., Selvamurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jagannathan M.K., Tata McGraw-Hill, New Delhi, 2012.
3 Biology for Engineers, Arthur T. Johnson, CRC Press, Taylor and Francis, 2011

Reference Books
1 Biology for Engineers, Sohini Singh and Tanu Allen, Vayu Education of India, New Delhi, 2014.
2 Biomimetics: Nature-Based Innovation, Yoseph Bar-Cohen, 1st edition, 2012, CRC Press.
3 Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies, D. Floreano and C. Mattiussi, MIT Press, 2008.
4 Bioremediation of heavy metals: bacterial participation, by C R Sunilkumar, N Geetha A C Udayashankar Lambert Academic Publishing, 2019.



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DATA VISUALIZATION LABORATORY

Course Code	21AIL481	Credits:	01
L:T:P:S	0:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	03

Course Objectives	
1	To Understand the importance of data visualization for business intelligence and decision making.
2	Know approaches to understand visual perception.
3	Learn about categories of visualization and application areas.
4	Familiarize with the data visualization tools.
5	Gain knowledge of effective data visuals to solve workplace problems.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply data visualization techniques to diverse datasets to create appropriate visualizations and to identify both positive and negative impacts of data-informed decision.	L3
CO2	Analyze the data visualization tools to represent and interpret the results of stories created using various features.	L4
CO3	Design interactive visualization by analysing the data visualization tools and libraries that allow users to explore and gain insights from data.	L4
CO4	Write clear and concisereport to document experimental procedures, observations, and results accurately, adhering to established standards.	L5

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	3	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	TFL Bus Safety: i. Create a bar chart on boroughs field to visualize the trend in the count. ii. Create a line chart for date of incidence for each month in a quarter, comment on possibilities and suitability of different charts for this timeline. iii. In above question, apply formatting to display the first letter of the month on X-axis. iv. Create tree maps of all the data fields except date & year and comment on significance of tree map. v. Create an interactive dashboard for the above data.	2	CIE+SEE+QUIZ	CO1,2,3,4



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2	GDP dataset: - Visualize the countries data given in the dataset with respect to latitude and longitude along with country name using symbol maps. - Create a bar graph to compare GDP of Belgium between 2006 – 2026. - Using pie chart, visualize the GDP of India, Nepal, Romania, South Asia, Singapore by the year 2010. - Visualize the countries Bhutan & Costa Rica competing in terms of GDP. - Create a scatter plot or circle views of GDP of Mexico, Algeria, Fiji, Estonia from 2004 to 2006. - Build an interactive dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
3	Customer Analysis dataset: - Create a choropleth map (fill the map) to spot the special trends to show the state which has the highest revenue. - Create a line chart to show the revenue based on the month of the year. - Create a bin of size 10 for the age measure to create a new dimension to show the revenue. - Create a donut chart view to show the percentage of revenue per region by creating zero access in the calculated field. - Create a butterfly chart by reversing the bar chart to compare female & male revenue based on product category. - Create a calculated field to show the average revenue per state & display profitable & non-profitable state. - Build a dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
4	Amazon Prime Dataset: - Create a Donut chart to show the percentage of movie and tv shows - Create a area chart to shows by release year and type. - Create a horizontal bar chart to show Top 10 genre. - Create a map to display total shows by country. - Create a text sheet to show the description of any movie/movies. - Build an interactive Dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
5	HR Dataset: - Create KPI to show employee count, attrition count, attrition rate, attrition count, active employees, and average age. - Create a Lollipop Chart to show the attrition rate based on gender category. - Create a pie chart to show the attrition percentage based on Department Category - Create a bar chart to display the number of employees by Age group, - Create a highlight table to show the Job Satisfaction Rating for each job role based on employee count. - Create a horizontal bar chart to show the attrition count for each Education field - Create multiple donut chart to show the Attrition Rate by Gender for different Age group. - Build a dashboard. Video Lectures:	2	CIE+SEE+QUIZ	CO1,2 ,3,4



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6	IPL Dataset: - Create a calculated field to display match winner for each season in KPI. - Find out orange cap holder by selecting the appropriate filter & display in KPI. - Find out purple cap holder by selecting the appropriate filter & display in KPI. - Find out the count of tournament sixes by selecting the appropriate filter & display in KPI. - Find out the count of tournament fours by selecting the appropriate filter & display in KPI. - Create a horizontal bar chart to display matches won based on toss decision. - Create a donut chart to display the winning percentage based on toss decision. - Create a duplicate sheet for the team winners, orange cap & purple cap to display the respective images by uploading the images in Tableau repository. - Use features of horizontal containers, blank sheet & build an interactive dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
7	Sales Profit Dataset: - Create KPI for sales profit and quantity using respective spark line. - Visualize sales & profit distribution by state (connecting with hexmap dataset). - Visualize average sales & profit by state along with respective state count, - Create a dynamic chart visualizing monthly states by segment. - Find out total sales by location & manager. - Build an interactive dashboard using the analysis.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
8	Car Sales Dataset: - Create a bar chart on Dataset. - Create a line chart - Create a tree map - Build an interactive dashboard which includes all the information & charts.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
9	Credit Card Complaint Dataset: - Create the following charts to track credit card complaints: - Total complaints along with rolling 12 months complaints with spark line chart. - Timely response with closed percentage progress bar. “In progress” along with its percentage – in progress spark line. - Create a dynamic chart to reflect trends as per the time duration (yearly, monthly, quarterly, weekly & daily). - Create a geographical density map with appropriate state & zip or pin code. - Create a filled map for the same with the colour gradient differentiation. - Sort top 10 issues based on its types. - Show data to display the company responses. - Create a calendar to show daily complaints. Use colour gradient to display. - Build an interactive dashboard which includes all the information	2	CIE+SEE+QUIZ	CO1,2 ,3,4



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	& charts.			
10	Visualizing ML Models using given Datasets	2	CIE+SEE+QUIZ	CO1,2 ,3,4

System Requirements:

1. System requirements are listed here under Tableau Desktop and Tableau Prep:
<https://www.tableau.com/products/techspecs>
2. The latest version of Tableau Desktop as well as Tableau Prep should be downloaded and installed from here: <https://www.tableau.com/tft/activation>

Text Books	
1	"Beautiful Visualization, Looking at Data Through the Eyes of Experts by Julie Steele, Noah Iliinsky"
2	"Information Dashboard Design: Displaying Data for At-a-glance Monitoring" by Stephen Few
3	Storytelling With Data: A Data Visualization Guide for Business Professionals" by Cole Nussbaumer Knaflic



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OOPS WITH JAVA

Course Code	21APL46	Credits:	01
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	03

Course Objectives	
1	Understand the syntax, semantics, features and applications of Java Programming Language.
2	Learn the core concepts and construct using Java.
3	Understand the advanced concept and creating Java application.
4	Learn the method of creating connection between Java and Database.
5	Design and build simple Java applications.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Make use of fundamentals of object-oriented programing structure and understand the basics of Java.	L3
CO2	Inspect the class, constructor, inheritance and polymorphism concepts of Java with real time applications.	L4
CO3	Analyze the different packages available in Java and implement them on the application	L4
CO4	Determine the process of creation of chatbots using Java AI	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	2	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Fundamentals of Java, Programming in Java, Tools available for Java programming, Building application in Java	2	CIE+SEE+QUIZ	CO1
2	Object oriented mechanism in Java, Class, Constructor, Inheritance, Polymorphism in Java, Access Specification	2	CIE+SEE+QUIZ	CO1 CO4
3	Interfaces, Packages, Exception handling in Java, Thread and Multithreading, Application development in Java	2	CIE+SEE+QUIZ	CO2 CO4



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4	String handling, String Constructors, String Operations, Character Extraction, String Comparison, Searching and Modifying Strings, StringBuffer, StringBuilder.	2	CIE+SEE+QUIZ	CO2 CO4
5	Java AIML(Artificial Intelligence Markup Language), Understanding NLP, Using the AIML Package, Creation of simple chatbots	2	CIE+SEE+QUIZ	CO1 CO4

Text Books	
1	Java_-The Complete Reference, 11th edition, Herbert Schildt, Mc Graw Hill Education, 2019.
2	Programming with Java, E. Balagurusamy.
3	Head First Java: A Brain-Friendly Guide, Third Edition (Grayscale Indian Edition)

Reference Books	
1	J2SE Core Java, A.R.Kishore Kumar.



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UNIVERSAL HUMAN VALUES

Course Code	21UH49	Credits:	01
L:T:P:S	0:1:0:0	CIE Marks : SEE Marks	50:50
Course Type	UHV	Exam Hours	03

Course Objectives	
1	Recognize the need for Value education harmony in self, family and society.
2	Understand the underlying the propositions of harmony in nature.
3	Verify the propositions and realize the implications of holistic understanding and professional ethics.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Recognize the importance of Value education.	L2
CO2	Understand the concepts of harmony in self.	L2
CO3	Develop skills for living in harmony and apply in professional life.	L3
CO4	Outline the principles of human values and examine its role in education	L2
CO5	Adapt to natural acceptance of human values build holistic understanding	L2
CO6	Test and verify the propositions on human values by self-exploration	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
1	Introduction to Value Education (3 hours) : Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education), Understanding Value Education, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Happiness and Prosperity – Current Scenario, Case studies, videos and real world examples	3	CIE+SEE+QUIZ	CO1 CO2
2	Harmony in the Human Being (3 hours) : Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self, Understanding Harmony in the Self, Harmony of the Self with the Body, Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	CO1 CO2



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3	Harmony in the Family and Society (3 hours) : Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship, 'Respect' – as the Right Evaluation, Other Feelings, Justice in Human-to-Human Relationship, Understanding Harmony in the Society, Vision for the Universal Human Order. Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	C03 C04
4	Harmony in the Nature/Existence (3 hours): Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfillment among the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence. Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	C03 C04
5	Implications of the Holistic Understanding: a Look at Professional Ethics (3 hours) Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession. Case studies, videos and real world examples.Presentation/Videos	3	CIE+SEE+QUIZ	C05 C06

Text Books	
1	A foundation course in HUMAN VALUES and Professional ethics; presenting a universal approach to value education- through self-exploration by, R R Gaur, R Sangal & GP Bagaria, Excel books Pvt. Ltd.
2	Professional Ethics & Human Values: Prof. D.R. Kiran, TATA Mc Graw Hill Education.

Reference Books	
1	Human Values: A. N. Tripathy (2003, New Age International Publishers)
2	Ethics in Engineering Mike W. Martin, Department of Philosophy, Chapman University and Roland Schinzinger, School of Engineering, University of California, Irvine.
3	Fundamentals of Ethics, Edmond G. Seebauer & Robert L. Barry, Oxford University Press



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ARTIFICIAL INTELLIGENCE

Course Code	21AI51	Credits	3
L:T:P:S	2:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	To develop the understanding of fundamentals and technological aspects of Artificial Intelligence.
2	To learn about basics of how to use Artificial Intelligence.
3	To understand and explain the various Real World Applications of Artificial Intelligence.
4	Develop an understanding of responsible AI practices and ethical guidelines.
5	Develop a solid understanding of the basic principles, concepts, and techniques of Artificial Intelligence.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Demonstrate a comprehensive understanding of the fundamental concepts and principles that basis artificial intelligence.	L1
CO2	Identify problems where artificial intelligence techniques are applicable.	L2
CO3	Apply AI algorithms to solve complex problems in various domains.	L3
CO4	Analyse real-world AI applications across various domains.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2														
CO3	3													
CO4		3												

Module	Topic	Hours	Assessment	CO
1	Module-1: Introduction and History: What is AI? , Foundation of AI, History of AI, State of the Art in AI Intelligent Agents: Agents and Environments, Good Behaviour: Concept of Rationality, Nature of Environments, Structure of Agents Solving Problem by Searching: Problem Solving Agents: Well-defined problems and solutions, Formulating problems Example Problems: Toy problems, Real-world problems	08	CIA,Quiz	C01
2	Module-2: Searching for solutions: Infrastructure for search algorithms, Measuring problem-solving performance, Adversarial Search and Games: Game Theory- Game Theory, Optimal Decision in Games, Heuristic Alpha-Beta Tree Search, Monte Carlo Tree Search	08	CIA,Quiz	CO2



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3	Module-3: Logical Agents: Knowledge Based Agents, The Wumpus World, Logic, Propositional Logic: A Very Simple Logic, propositional Theorem, Proving, Effective Propositional Model Checking, Agents Based on Propositional Logic	08	CIA, Quiz	CO3
4	Module-4: Knowledge Representation: Ontological Engineering, Categories and Objects, Events, Mental Events and Mental Object	08	CIA, Quiz	CO4
5	Module-5: The Ethics of AI- Introduction, Unfair and Illegal Discrimination, Privacy, Surveillance Capitalism, Right to Life, Liberty and Security of Persons, AI for Good and the SDGs	08	CIA, Quiz	CO5

Text Books	
1	Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell A First Course in Artificial Intelligence by Deepak Khemani
2	Ethics of Artificial Intelligence Case Studies and Options for Addressing Ethical Challenges by Bernd Carsten Stahl · Doris Schroeder · Rowena Rodrigues

Reference Books	
1	NCERT Introduction to AI—Curate with the Support of Intel https://cbseacademic.nic.in/web_material/Curriculum21/publication/secondary/Class10_Facilitator_Handbook.pdf



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Department of Artificial Intelligence and Machine Learning

ADVANCED MACHINE LEARNING

Course Code	21AI52	Credits	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	3

Course Objectives

1	To introduce the fundamental techniques and principles of Neural Networks.
2	To study the different models in ANN and their applications.
3	To familiarize deep learning concepts with Convolutional Neural Network case studies.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand a wide variety of learning algorithms and principles of Deep Learning	L1
CO2	Apply ensemble model and recommendation system to real world scenarios	L3
CO3	Analyze the concepts of ANN, CNN and RNN models	L4
CO4	Implement Sequence-to-Sequence Models for simple applications	L3
CO5	Document the observations of implementing various models along their performance	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4					3									
CO5					3				2	2				

Module	Topic	Hours	Assessment	CO
1	Time Series Forecasting: Stationary of Data, Time Series Decomposing, Moving average, Exponential Smoothing, Holt-Winters Algorithm, ARIMA model and Components	08	CIA,Quiz	C01
2	Ensemble Technique-1: Bagging, Boosting, Voting, GBM (Gradient Boosting Machines), Evaluating Ensemble classifier, Random Forest Algorithm, Gradient Boosting Algorithm, XG- Boost. Recommendation Systems: Algorithmic ingredients of a Recommendation Engine, Dimensionality Reduction.	08	CIA,Quiz	CO2
3	Introduction to Neural Network: Deep Feedforwd Networks-What is neural network?, How neural networks works? Gradient descent, Stochastic Gradient descent, Perceptron, Multilayer Perceptron, Back Propagation, ANN	08	CIA,Quiz	CO3
4	Convolutional Networks: The Convolution Operation, Motivation, Pooling Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Random or Unsupervised Features, The Neuro scientific Basis for Convolutional Networks, Convolutional Networks and the History of Deep Learning	08	CIA,Quiz	CO3



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	Implementation of CNN for real world problemsEvents, Mental Events and Mental Object			
5	Introduction to Recurrent Neural Networks (RNN) What are Recurrent Neural Networks(RNNs)?, Understanding a Recurrent Neuron in Detail, Long Short-Term Memory(LSTM),Back propagation Through Time(BPTT), Implementation of RNN in Keras. Sequence-to-Sequence Models: Importing all of the dependencies, Defining the hyper parameters, Building a simple deep neural network, Pooling, Drop out technique, Data augmentation.	08	CIA,Quiz	CO3 CO4

Text Books	
1	Deep Learning , Ian Goodfellow, Yoshua Bengio and Aaron Courville, - MIT Press , ISBN 9780262035613, Published: November 18, 2016
2	Data Science from Scratch, First Principles with Python by Joel Grus, Deep Learning Ian Goodfellow, Yoshua Bengio, Aaron Courvill.

Reference Books	
1	Forecasting: Principles and Practice:Hyndman, R. & Athanasopoulos, G., 2021, 3rd ed. OTexts.



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COMPUTER NETWORKS AND COMMUNICATION

Course Code	21AI53	Credits	3
L:T:P:S	2:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	Acquire Profound Knowledge in Routing, Internetworking, and Congestion Control
2	Analyze and Compare Diverse Network Architectures and Topologies
3	Analyze and Compare Network Models, Emphasizing TCP/IP Protocol Suite and OSI Model
4	Analyze Packet Routing and Conduct Case Studies

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Analyze and gain the knowledge of routing, internetworking, congestion control and quality of service.	L4
CO2	Analyze and compare various network architectures and topologies.	L4
CO3	Analyze and compare various network models, focusing on the TCP/IP protocol suite and OSI Model.	L4
CO4	Analyze Packet Routing Protocol and case studies.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1		3												
CO2		3												
CO3		3												
CO4		3												
CO5		3												

Module	Topic	Hours	Assessment	CO
1	Introduction: Networks, Network types. Network Models: TCP / IP protocol suite, Addressing, The OSI Model. Transmission Modes: Parallel Transmission and Serial Transmission. Link Layer: Data Link Control (DLC): DLC Services, Data Link Layer Protocols. Media Access Control (MAC): Random Access: CSMA/CD, CSMA/CA.	08	CIA, Quiz	CO3, CO2
2	Network layer design issues: Store and Forward packet Switching, Services Provided to the Transport Layer Implementation of Connectionless Service, Implementation of Connection-Oriented Service, Comparison of Virtual Circuit and Datagram Subnets; Routing algorithms: Shortest Path Routing, flooding distance vector routing, link state routing , Hierarchical Routing Broadcast Routing, and Multicast Routing.	08	CIA, Quiz	CO1, CO4
3	Congestion Control Algorithms: General Principles of Congestion Control, Congestion Prevention Policies, and Congestion Control in Virtual- Circuit Subnets, Congestion Control in Datagram Subnets, Load Shedding, Jitter Control; Quality Of Service: Requirements,	08	CIA, Quiz	CO1



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	Techniques for Achieving Good Quality of Service Integrated Services Differentiated Services.			
4	The Network Layer in the Internet: The IP Protocol, IP Addresses, Internet Control Protocols, OSPF- Interior Gateway Routing Protocol, BGP- Exterior Gateway Routing Protocol, IPv6. The Transport Service: Services Provided to the upperlayer.. The Internet Transport Protocols: Introduction to UDP, RPC, RTCP, Introduction to TCP. The TCP Service Model.	08	CIA, Quiz	CO3, CO4
5	The TCP Protocol: TCP protocol, TCP Segment Header, TCP Connection Establishment, TCP Connection Release. TCP Transmission Policy, TCP Congestion Control, TCP Timer Management. Application Layer: World Wide web and HTTP, FTP, Electronic Mail, Telnet.	08	CIA, Quiz	CO4

Text Books	
1	Data Communications and Networking, Behrouz A Forouzan, 5th Edition, 2013, Tata McGraw-Hill, ISBN – 9781259064753.

Reference Books	
1	Computer Networks, Andrew S Tanenbaum, 5th Edition, 2014, Pearson Education; ISBN – 978-81- 7758-165-2.
2	Computer Networking, a Top-Down Approach, James Kurose and Keith Ross, 6th Edition, 2013, ISBN- 13: 978-0-13-285620-1.
3	Data and Computer Communications, William Stallings, 8th Edition, 2009, Pearson Education, ISBN- 13: 978-0131392052.



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INTRODUCTION TO ROBOTICS

Course Code	21AI54	Credits	03
L:T:P:S	2:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	To study operations and components of robotics.
2	To study how different types of sensors and actuators are implemented in a microcontroller.
3	To learn robot programming.
4	To learn how robot implemented in different areas.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Elucidate the fundamentals of robotics and its components.	L1
CO2	Identify the precise instruction that will enable a robot to function.	L2
CO3	Analyze the robotic technology used in the different domains.	L4
CO4	Design Robotic system that demonstrates a real time capability considering data integration and sensor fusion technologies.	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2														
CO3		3												
CO4	3													

Module	Topic	Hours	Assessment	CO
1	INTRODUCTION TO ROBOTS: automation and its types, History and evaluation of robotics, Definition of robots, Robotic Manipulators, Types of robots, Generations of robots, Laws of robotics, Classification of robots and its applications, Classification of robots and its applications, Specifications of robots	08	CIA,Quiz	CO1 CO3
2	ROBOT ANATOMY: Anatomy of robots, Drive systems, Actuators and power Transmission system, Types of drives and its application, Hydraulic drives, Pneumatic drives, Electric drives, hybrid drives, Robot activation and feedback components.	08	CIA,Quiz	CO1 CO2, CO3
3	SENSORS IN ROBOTICS : Touch Sensors, Tactile Sensors, Proximity and Range sensors, Sensor Based Systems, Force sensor, Light sensors, Pressure sensors, Ultrasonic sensors, Infrared sensors, Pots, Encoders, Position & Velocity Sensors, Materials used for robot design and its Properties, Transmission devices in robots and its types, End effectors, Types of end effectors.	08	CIA,Quiz	CO1 CO2, CO3
4	ROBOTCONTROLLERS AND PROGRAMMING: Classification of tools and grippers, Types Tools and grippers actuations, Robot Brain, Controller and its types, Need for controller in robotics, Robotic simulation, Robot software, Robot programming and the languages.	08	CIA,Quiz	CO2, CO3



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5	ROBOT APPLICATIONS: Industrial applications of robots, Medical, Household, Entertainment, space, Underwater, Defense, Environmental and economic issues in robot applications, Advantages and disadvantages of Robotization.	08	CIA, Quiz	CO1 CO4
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Text Books

1	Groover M.P., “Industrial Robotics -Technology Programming and Applications”, McGraw Hill, 2001.
2	D.T.C.Manjunath, “Fundamentals of Robotics”, Nandu Publishers, 5th edition, India, 2005
3	Robotics_control_sensing_vision and Intelligence by Fu, Gonzalez and Lee.

Reference Books

1	Murphy Robin R,” Introduction to AI Robotics”, MIT Press, 2000. no reprint available
2	Siegwart R and Nourbakhsh I.R, “Introduction to Autonomous mobile Robots”, Prentice Hall India, 2005. no reprint available
3	Elaine and Kevin Knight, “Artificial Intelligence”, Second Edition



ADVANCED MACHINE LEARNING WITH LABORATORY

Course Code	21AI52	Credits	1
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	To analyze and implement time series analysis on suitable applications.
2	Analyze and Design Ensemble techniques for handling real data.
3	Analyze and implement concepts Neural Networks and Deep Networks

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Design and Implement Time Series Analysis to suitable applications.	L4
CO2	Design and Implement various Ensemble Models and recommendation on different datasets	L4
CO3	Design and Implement the basic concepts of neural network model and design the same	L4
CO4	Design and Implement Convolution neural networks to Real Life Problems.	L4
CO5	Design and Implement Recurrent Neural Networks to Real Life Problems	L4
CO6	Evaluate the performance of learning models	L5

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1		3												
CO2		3												
CO3		3												
CO4		3												
CO5		3												
CO6				3										

Module	Topic	Hours	Assessment	CO
1	Simulate the observations using ARIMA model in time series analysis for the given dataset	02	CIA,Quiz	CO1, CO6
2	Apply: Random Forest classifier on: i) Given Dataset, Draw Linearly separable decision boundary for the generated dataset. ii) Randomly generated dataset using package library [MAKEMOON], Draw Non-linearly separable decision boundary for the generated dataset.	02	CIA,Quiz	CO2, CO6
3	Apply: SVM classifier on: i) Iris Dataset, Draw Linearly separable decision boundary for the generated dataset. ii) Randomly generated dataset using package library [MAKEMOON], Draw Non-linearly separable decision boundary for the generated dataset	02	CIA,Quiz	CO2, CO6
4	Implementation of simple recommendation system on the given dataset	02	CIA,Quiz	CO2, CO6



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5	Demonstrate a) Usage of Sigmoid activation function in artificial neural network b) Identification of face using opencv library.	02	CIA, Quiz	CO3, CO6
6	Using Keras and Tensor flow framework i) Load the Pima_indians_diabetes dataset ii) Design a two-layer neural network with one hidden layer and one output layer a. Use Relu activation function for the hidden layer b. Use sigmoid activation function for the output layer iii) Train the designed network for Pima_indians_diabetes iv) Evaluate the network Generate Predictions for 10 samples	02	CIA, Quiz	CO3, CO6
7	Using Keras and tensor flow network Load the mnist image dataset ii) Design a two-layer neural network with one hidden layer and one output layer a. Use CNN with Leaky Relu activation function for the hidden layer b. Use sigmoid activation function for the output layer iii) Train the designed network for mnist dataset	02	CIA, Quiz	CO4, CO6



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COMPUTER NETWORKS LABORATORY

Course Code	21AIL55	Credits	1
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	Illustrate topology formation and working of stop-and-wait protocol.
2	Examine working of network connections and LAN working.
3	Illustrate the working of error detection and finding the shortest path using Bellman- ford algorithm.
4	Examine RSA algorithm used for security and Congestion Control concepts.
5	Demonstrate socket programming using TCP protocol

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Topology formation and working in Cisco packet tracer	L3
CO2	Examine point -to -point network and LAN working (Wired and Wireless) using NS2	L3
CO3	Demonstrate the working of stop-and-wait protocol in NS2	L4
CO4	Illustrate the working of error detection using CRC and finding the shortest path using Bellman- ford algorithm in Java	L3
CO5	Examine RSA algorithm used for security and Congestion Control by Leaky bucket in Java	L3
CO6	Demonstrate socket programming using TCP protocol in Java.	L3
Co7	Examine point -to -point network and LAN working (Wired and Wireless) using Cisco packet tracer	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3				3									
CO2	3				3									
CO3		3												
CO4	3													
CO5	3													
CO6	3													
CO7		3			3									

Module	Topic	Hours	Assessment	CO
1	Implement Ring topology and Bus topology operation using Cisco Packet Tracer.	02	CIA,Quiz	CO1
2	Implement three nodes point - to - point network with duplex links between them using Cisco packet tracer.	02	CIA,Quiz	CO7



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3	Implement an Ethernet LAN using n nodes and set multiple traffic nodes and plot congestion window for different source /destination.	02	CIA,Quiz	CO2
4	Write a NS2 script to implement the operation of Stop and Wait protocol	02	CIA,Quiz	CO3
5	Implement simple ESS and with transmitting nodes in wire-less LAN by simulation and determine the performance with respect to transmission of packets.	02	CIA,Quiz	CO3
6	Write a program for error detecting code using CRC-CCITT (16-bits).	02	CIA,Quiz	CO4
7	Write a program to find the shortest path between vertices using Bellman- Ford algorithm.	02	CIA,Quiz	CO4
8	Write a Java program for congestion control using leaky bucket algorithm.	02	CIA,Quiz	CO5
9	Write a Java program to Implement RSA algorithm to encrypt and decrypt the data while sending it and decrypt while receiving.	02	CIA,Quiz	CO5
10	Write a Java program to Implement TCP/IP sockets, write a client - server program to make the client send the file name and to make the server send back the contents of the requested file if present.	02	CIA,Quiz	CO6



RESEARCH METHODOLOGY AND IPR

Course Code	21RM56	Credits	2
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	2

Course Objectives

1	To identify the importance of Research.
2	To understand the research methodology, tools and techniques.
3	To interpret the results of Research and Propose recommendations.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Outline the importance of conducting research, and Intellectual property rights	L1
CO2	Categorize the different tools used for conducting research	L2
CO3	Compare various techniques used in defining and executing a research problem	L3
CO4	Prepare a flow diagram to define and conduct research for a given problem	L4
CO5	Deduce a methodology to address a given problem starting from defining it to reporting	L6

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2														
CO3	3													
CO4		3												
CO5		3								2				

Module	Topic	Hours	Assessment	CO
1	Meaning, Objectives and Characteristics of research - Research methods Vs Methodology - Types of research - Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical - Research approaches, Significance of Research, Research process - Criteria of good research.	04	CIA, Quiz	CO 1,4,5
2	Defining the research problem - Selecting the problem - Necessity of defining the problem - Techniques involved in defining the problem. Place of literature review in research, procedure for reviewing the literature, search for existing literature, develop a theoretical framework, and develop a Conceptual framework.	04	CIA, Quiz	CO 3,4,5
3	Research design-Meaning of Research Design, need for research design, features of a good design, Important concepts- Dependent and independent variables, Extraneous variable, Control, Confounded relationship, Research hypothesis, Experimental and non-experimental hypothesis-testing research, Experimental and control groups, Treatments, Experiment, Different research designs. Methods of Data Collection: Primary data, secondary data, collection of primary data-observation method, interview method, questionnaires, collection of	06	CIA, Quiz	CO 3,4,5



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	data through schedules, collection of secondary data			
4	Sampling design - Steps in sampling design - criteria of selecting a sampling procedure Characteristics of a good sample design - Types of sample designs-Non-probability sampling, Probability sampling, how to select a random sample, random sample from an infinite universe. Testing of hypotheses - Basic concepts - Procedure for hypotheses testing flow diagram for hypotheses testing Basic concepts concerning testing of hypotheses- null hypothesis and alternative hypothesis, level of significance, type I and type II errors, two tailed and one tailed tests, procedure for hypothesis testing (No Numerical)	06	CIA, Quiz	CO 3,4,5
5	IPRs- Invention and Creativity- Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs)- A brief summary of: Patents, Copyrights, Trademarks, Industrial Designs- Integrated Circuits- Geographical Indications- Establishment of WIPO- Application and procedures, Patenting Procedure. Interpretation and report writing - Techniques of interpretation - Structure and components of scientific reports - Different steps in the preparation- Layout, structure and language of the report - Illustrations and tables - Types of report - Technical reports and thesis	06	CIA, Quiz	CO 1,4,5

Text Books	
1	Research Methodology: A Step-by-Step Guide for Beginners ; Author, Ranjit Kumar ; Publisher, SAGE Publications, 2010 ; ISBN, 1446244776
2	"Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" by John W. Creswell and J. David Creswell

Reference Books	
1	Sampling techniques ; Author: William G. Cochran (Author) ; Edition: 3d ed View all formats and editions ; Publisher: Wiley, New York, NY, 1977
2	" Understanding intellectual property law :Ochoa, Tyler T., author.; Ghosh, Shubha, author.; LaFrance, Mary, 1958- author. 2020
3	"Basic Business Statistics: Concepts and Applications" by Mark L. Berenson, David M. Levine, and Timothy C. Krehbiel



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ENVIRONMENTAL STUDIES

Course Code	21ES57	Credits	1
L:T:P:S	1:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	HSMC	Exam Hours	2

Course Objectives

1	Develop sense of responsibility about the environment, natural resources, and their conservation
2	Understand the concept of various components of environment, their pollution and their remediation methods.
3	To study global environmental issues

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Discuss the components of environment	L2
CO2	Discuss the various sources of energy.	L2
CO3	Identifications of various types of Environmental pollutions and suggestions of remedial measures.	L2
CO4	Study of various wastes produced from society and their disposal methods.	L2
CO5	Identify the major challenges in environmental issues and suggest possible solutions.	L2
Co6	Understand global Environment problems.	L2

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1								3				2		
CO2								3				2		
CO3								3				2		
CO4								3				2		
CO5								3				2		

Module	Topic	Hours	Assessment	CO
1	Introduction: Environment - Components of Environment Ecosystem: Types & Structure of Ecosystem. Water resources: Availability, quality, water borne & water induced diseases	02	CIA,Quiz	CO1
2	Energy: Different types of energy, Conventional sources & Non-Conventional sources of energy Solar energy, Hydro electric energy, Wind Energy, Nuclear energy, Biomass & Biogas, Fossil Fuels, Hydrogen as an alternative energy.	02	CIA,Quiz	CO2
3	Environmental Pollution -Water Pollution, Noise pollution, Land Pollution- sources, effects and remedies. Air Pollution: Definition, Effects and remedies	03	CIA,Quiz	CO3



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4	Solid Waste Management: E-Waste, Hazardous waste & Biomedical Waste -Sources, Characteristics & Disposal methods, Radioactive waste, construction and demolition debris, and mining waste.	03	CIA, Quiz	CO4 CO5
5	Global Environmental Issues: Global Warming, Acid rain & Ozone layer depletion, controlling measures, heat island effect, Waste to energy concepts.	02	CIA, Quiz	CO6

Text Books

1	S.M. Prakash, "Environmental Studies", Elite Publishers Mangalore, 2007
2	Benny Joseph (2005), "Environmental Studies", Tata McGraw – Hill Publishing Company Limited.
3	R. J. Ranjit Daniels and Jagadish Krishnaswamy, (2009), "Environmental Studies", Wiley India Private Ltd., New Delhi.
4	R Rajagopalan, "Environmental Studies – From Crisis to Cure", Oxford University Press, 2005.

Reference Books

1	Raman Sivakumar, "Principals of Environmental Science and Engineering", Second Edition, Cengage learning Singapore, 2005
2	P. Meenakshi, "Elements of Environmental Science and Engineering", Prentice Hall of India Private Limited, New Delhi, 2006
3	Aloka Debi, "Environmental Science and Engineering", Universities Press (India) Pvt. Ltd. 2012 4.G.Tyler Miller Jr., "Environmental Science – working with the Earth", Eleventh Edition, Thomson Brooks /Cole, 2006



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FULL STACK DEVELOPMENT LAB

Course Code	21AIL584	Credits	1
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	3

Course Objectives

1	To become knowledgeable about the most recent web development technologies
2	Idea for creating two tier and three tier architectural web applications
3	Design and Analyse real time web applications
4	Constructing suitable client and server side applications.
5	To learn core concept of both front end and back end programming.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Applying JavaScript fundamentals to solve real-world problems.	L3
CO2	Develop functional React applications	L6
CO3	Building Full-Stack Applications with ExpressJS, NodeJS, and MongoDB:	L6
CO4	Developing complete web applications using the MERN Stack.	L6

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3													
CO2		3												
CO3					3				2	2				
CO4					3				2	2				

Module	Topic	Hours	Assessment	CO
1	Introduction to JavaScript (Overview, Syntax, Placement) and Variables	02	CIA, Quiz, Project	CO1
2	JavaScript Operators ,Conditional Statements and JavaScript Loops	02	CIA, Quiz	CO1
3	JavaScript Functions	02	CIA, Quiz	CO1
4	Introduction to React, React Features, and Setting up React	02	CIA, Quiz	CO2
5	React Events and Forms (2 hours)	02	CIA, Quiz	CO2
6	React Lists, Keys, and Refs	02	CIA, Quiz	CO2
7	Forms and Form Validation in React	02	CIA, Quiz	CO2
8	Introduction to ExpressJS and Setting Up a Server	02	CIA, Quiz	CO3
9	Routing in ExpressJS	02	CIA, Quiz	CO3
10	Handling Form Data and Connecting to Databases	02	CIA, Quiz, Project	CO3

Text Books



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1	"Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node" by Vasana Subramanian, Publication Year: 2017.
2	"Learning React: Modern Patterns for Developing React Apps" by Alex Banks and Eve Porcello, Year: 2017 O'Reilly Banks Media and Eve Porcello.
3	MongoDB: The Definitive Guide" by Kristina Chodorow and Shannon Bradshaw, Publication Year: 2013 (2nd Edition), Publisher: O'Reilly Banks Media.

Reference Books	
1	Full-Stack JavaScript Development by Eric Bush.
2	Mastering Full Stack React Web Development Paperback – April 28, 2017 by Tomasz Dyl , Kamil Przeorski , Maciej Czarnecki
3	Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon

NATURAL LANGUAGE PROCESSING

Course Code	21AI61	Credits	03
L:T:P:S	2:2:0: 0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	Understand the importance of natural language processing (NLP) and its applications in text analysis and generation.
2	Develop practical skills in conducting NLP experiments for tasks such as information retrieval and language generation.
3	Gain proficiency in implementing NLP algorithms for building language generation systems and machine translation models.
4	Learn to apply information retrieval techniques and lexical resources effectively in NLP tasks.
5	Explore advanced NLP topics and apply them in real-world scenarios to solve language processing challenges

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Identify the importance of natural language processing and tag a given text with basic language features	L2
CO2	Apply word-level and syntactic analysis techniques in NLP tasks.	L3
CO3	Analyze semantic structures and discourse in natural language understanding.	L4
CO4	Analyze machine translation approaches, considering the characteristics of Indian languages and translation involving these languages.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	OVERVIEW AND LANGUAGE MODELING Overview: Origins and challenges of NLP, Language and Grammar, Processing Indian Languages, NLP Applications, Information Retrieval.]Language Modeling: Various Grammar- based Language Models, Statistical Language Model, N-Grams Model, Minimum Edit Distance	08	CIA,Quiz	CO1,
2	WORD LEVEL AND SYNTACTIC ANALYSIS Word Level Analysis: Regular Expressions, Finite-State Automata, Morphological Parsing, Spelling Error Detection and correction, Words and Word Classes, Part-of-Speech Tagging. Syntactic Analysis: Context-free Grammar, Constituency, Parsing, Probabilistic Parsing.	08	CIA,Quiz	CO2
3	SEMANTIC ANALYSIS AND DISCOURSE PROCESSING Semantic Analysis: Meaning Representation, Lexical Semantics, Ambiguity, Word Sense Disambiguation. Discourse Processing: Cohesion, Reference Resolution, Discourse Coherence and Structure.	08	CIA,Quiz	CO3



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4	NATURAL LANGUAGE GENERATION AND MACHINE TRANSLATION Natural Language Generation: Architecture of NLG Systems, Generation Tasks and Representations, Application of NLG, Machine Translation: Problems in Machine Translation, Characteristics of Indian Languages, Machine Translation Approaches, Translation involving Indian Languages.	08	CIA, Quiz	CO4
5	INFORMATION RETRIEVAL AND LEXICAL RESOURCES Information Retrieval: Design features of Information Retrieval Systems, Information Retrieval Models, Classical Information Retrieval Models, Non-classical models of IR, Alternative Models of IR, Evaluation of the IR System, Lexical Resources: Word Net, Frame Net, Stemmers, Part-of Speech Tagger, Research Corpora.	08	CIA, Quiz	CO5

Text Books	
1	Natural Language Processing and Information Retrieval, Tanveer Siddiqui, U.S.Tiwary, 1 st Edition Oxford University press 2019.

Reference Books	
1	Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Daniel Jurafsky and James H Martin, 2 nd Edition Prentice Hall 2019
2	Natural Language Understanding, James Allen, 2 nd Edition Benjamin /Cummings, Publishing Company 2017.



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APPLIED BIG DATA AND CLOUD COMPUTING

Course Code	21AI62	Credits	04
L:T:P:S	3:0:2: 0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	3

Course Objectives

1	To understand the need of Big Data, challenges and different analytical architectures
2	Installation and understanding of Hadoop Architecture and its eco systems
3	Processing of Big Data with Advanced architectures like Spark.
4	Describe graphs and streaming data in Spark

Course Outcomes

**Blooms
Taxonomy(highest)**

CO1	Demonstrate a solid understanding of the principles, characteristics, and challenges of big data and cloud computing.	L1
CO2	Apply cloud computing concepts to effectively deploy, manage, and scale big data infrastructure and resources.	L3
CO3	Examine practical scenarios and real-life examples where the integration of Big Data analytics and cloud computing technologies has been applied to address various challenges.	L4
CO4	Develop analytical skills to systematically analyze and select optimal tools based on problem analysis.	L4
CO5	Implementing, and deploying a big data analytics solution on a cloud computing platform	L4
CO6	Create simple software programs and web applications, as well as effectively utilize programming tools and environments	L6

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3					3									
CO4		3												
CO5		3												
CO6					3				2	2				



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Module	Topic	Hours	Assessment	CO
1	INTRODUCTION TO CLOUD COMPUTING Introduction, Cloud Computing at a Glance, Defining a Cloud, Cloud Computing Reference Model, Characteristics and Benefits, Challenges Ahead, Historical Developments, Distributed Systems, Computing Platforms and Technologies. VIRTUALIZATION: Introduction, Characteristics of Virtualized Environments, Taxonomy of Virtualization Techniques, Execution Virtualization, Virtualization and Cloud Computing, Pros and Cons of Virtualization.	08	CIA, Quiz	C01, C02
2	CLOUD COMPUTING ARCHITECTURE: Introduction, Cloud Reference Model, Architecture, Infrastructure / Hardware as a Service, Platform as a Service, Software as a Service, Types of Clouds, Public Clouds, Private Clouds, Hybrid Clouds, Community Clouds, Economics of the Cloud, Open Challenges, Cloud Definition, Cloud Interoperability and Standards Scalability and Fault Tolerance, Security, Trust and Privacy. CLOUD APPLICATION PROGRAMMING AND THE ANEKA PLATFORM Framework Overview, Anatomy of the Aneka Container, From the Ground Up: Platform Abstraction Layer, Fabric Services, foundation Services, Application Services, Building Aneka Clouds, Infrastructure Organization, Logical Organization.	08	CIA, Quiz	C01, C02, C04
3	DATA INTENSIVE COMPUTING: What is Data-Intensive Computing? Characterizing Data-Intensive Computations, Challenges ahead, Historical perspective. HADOOP DISTRIBUTED FILE SYSTEM BASICS: Hadoop Distributed File System Design, Feature, HDFS Components, HDFS User Commands, HDFS Web GUI, Using HDFS in Programs.	08	CIA, Quiz	C01, C03, C04
4	RUNNING EXAMPLE PROGRAMS AND BENCHMARKS: Running MapReduce Examples, Running Basic Hadoop Benchmarks. HADOOP MAPREDUCE FRAMEWORK: The MapReduce Model, MapReduce Parallel Data Flow, Fault Tolerance and Speculative Execution. MAPREDUCE PROGRAMMING : Compiling and Running the Hadoop, WordCount Example , Using the Streaming Interface, Using the Pipes Interface.	08	CIA, Quiz	C05, C06
5	SPARK FRAMEWORK : Overview of Spark, Hadoop vs Spark, Cluster Design, Cluster Management, performance, APPLICATION PROGRAMMING INTERFACE (API): Spark Context, Resilient Distributed Datasets, Creating RDD, RDD Operations, Saving RDD, Lazy Operation, Spark Jobs.	08	CIA, Quiz	C01, C04, C05, C06

Text Books

1	. Douglas Eadline, "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem", 1 st Edition, Pearson Education, 2016. ISBN-13: 978-9332570351.
2	Anil Maheshwari, "Data Analytics", 1st Edition, McGraw Hill Education, 2017. ISBN-13: 978-9352604180.
3	"Mastering Cloud Computing Foundations and Applications Programming", Rajkumar Buyya Christian Vecchiola, S. Thamarai Selvi, Morgan Kaufmann is an imprint of Elsevier 225 Wyman Street, Waltham, MA 02451, USA.



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Reference Books	
1	Frank Pane, —Hands On Data Science and Python Machine Learning, Packt Publishers, 2017.
2	Nick Pentreath, Machine Learning with Spark, Packt Publishing, 2015
3	Seema Acharya, Subhashini Chellapan, —Big Data and Analytics, Wiley, 2015.
4	Big Analytics & Cloud Computing, First Edition 2021.



MACHINE LEARNING OPERATIONS

Course Code	21AI63	Credits	3
L:T:P:S	2:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	To understand the fundamentals of ML Operations
2	To learn about the basics tools of MLOps
3	To build containers for the ML pipeline
4	To apply end-to-end MLOps workflow

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand the lifecycle of a Machine Learning model	L1
CO2	Leverage the power of MLOps to productionalise Machine Learning models at scale	L2
CO3	Gain the best practices for putting Machine Learning models in production	L3
CO4	Get some insights on how to choose your perfect MLOps stack	L3
CO5	Understand the significance of Git and Git Operations	L2

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2														
CO3	3													
CO4	3													
CO5														

Module	Topic	Hours	Assessment	CO
1	Intro to MLOps What is MLOPs? Machine Learning Life Cycle, MLOPs Motivation: High-level view, MLOPs challenges MLOPs Components, MLOPs maturity models Detailed MLOPs and stages –Versioning, Testing, Automation (CI/CD), Reproducibility, Deployment, Monitoring. MLOPs tool ecosystem, Different Roles involved in MLOPs ML Pipeline Vs Automation Pipeline, Introduction to CI and CD Steps involved in the CI/CD implementation in ML lifecycle and workflow.	08	CIA, Quiz	CO1



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2	Intro to Git and GitHub Overview of Git, Understanding branching strategies and repo, Standard GIT branching strategies (development, feature, bug, release, UAT), Practicing important Git commands (LINUX) Intro to github, Create github account, Github action overview	08	CIA, Quiz	C02
3	Intro to Docker and Kubernetes Docker Foundation Kubernetes Overview, Kubernetes Architecture, Kubernetes Resources, Kubernetes Deployment Strategy	08	CIA, Quiz	C03
4	Intro to Azure DevOps ML Software serving processes, Adoption of DevOps practices in ML world, Complexities involved in MLOPs (Difference for regular Software Engineering), Intro to CI / CD for Machine Learning, Tools for CI / CD for ML, Intro to Cloud Environment MLOps Spinning resource with Azure Cloud, Automation with CI/CD, YAML coding	08	CIA, Quiz	C04
5	Azure MLOPs Azure Machine learning studio, Azure ML components Fully automated end-to-end CI/CD ML pipelines, Customer Churn Prediction, Serialize the model, Introduction to Streamlit – Streamlit HTML CSS, Create a basic Streamlit app, Web hosting for the app, Interacting with ML app with Get/POST methods, Introduction to Model Monitoring, Post deployment challenges	08	CIA, Quiz	C05

Text Books	
1	Practical MLOps, by Noah Gift, Alfredo Deza, Released September 2021 Publisher(s): O'Reilly Media, Inc., ISBN: 9781098103019

Reference Books	
1	Introducing MLOps, by Mark Treveil, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, Lynn Heidmann, Released November 2020, Publisher(s): O'Reilly Media, Inc. ISBN: 9781492083290
2	ML Ops: Operationalizing Data Science, by David Sweenor, Dev Kannabiran, Thomas Hill, Steven Hillion, Dan Rope and Michael O'Connell, O'Reilly Media, 2021
3	Building Machine Learning Pipelines, by Hannes Hapke, Catherine Nelson, O'Reilly Publications, 2021



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IOT AND 5G

Course Code	21AI643	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To provide skills for designing and analyzing IOT Concepts.
2	To understand fundamentals and architecture of internet of things.
3	To enable students to work on IoT Physical Devices.
4	To provide skills to work towards solution of real-life problems.
5	To build an embedded system using Raspberry Pi.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand the foundations, enabling technologies and devices of IoT	L2
CO2	Utilize Python programming skills to devise an IoT ecosystem employing Raspberry Pi and Arduino.	L3
CO3	Analyze the architecture and role of physical and cloud servers in IoT environment	L4
CO4	Analyze the vulnerabilities and security challenges in IoT	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	Introduction to Internet of Things – Definition and Characteristics of IoT, Physical & Logical Design of IoT, IoT Communication Models. IoT enabling Technologies – Wireless Sensor Networks, Cloud Computing, Big data analytics, Communication protocols, Embedded Systems, IoT Levels and Templates. Examples of Domain Specific IoTs – Health, City, Retail, Environment	08	CIA, Quiz	CO1
2	Introduction to Sensors: Data Acquisition- Sensors, Signals, and Systems, Sensor Classification, Units of Measurements. Sensor Characteristics - Accuracy, Calibration, Calibration Error, Hysteresis, Nonlinearity, Saturation, Special Properties, Dynamic Characteristics.	08	CIA, Quiz	CO1



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	Study Physical Characteristics Sensors- Temperature, capacitance, piezoelectric effect, strain etc. Sensors based on these physical characteristics			
3	IoT Physical Devices and Endpoints – Introduction to Raspberry PI- Interfaces (serial, SPI, I2C), Interfacing Sensors – temperature, LED, switch Programming – Language Overview, Python Packages of interest for IOT, Python program with Raspberry PI with focus of interfacing external gadgets, controlling output, and reading input from pins.	08	CIA, Quiz	CO1 CO2
4	IOT Physical Servers and Cloud Offerings – WAMP: Autobahn for IOT; Django: Python WEB application framework; Architecture; database configuration; Defining model, view and template; Amazon WEB services for IOT, RESTful API Case Studies Illustrating IOT Design – Any two of the four applications: Home automation, Cities, Environment, Agriculture	08	CIA, Quiz	CO3
5	Building IOT Systems using Arduino – Arduino overview and port, Use of ports for connecting an IO device, Controlling an LED with Arduino, Connecting sensors (temperature) to Arduino, Controlling actuator to Arduino IoT security challenges and requirements: Vulnerabilities and Threats for IoT – Attack surface areas, Standards and practices for IoT security – MQTT security, OWASP IoT Security.	08	CIA, Quiz	CO2 CO4

Text Books

1	1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015.
2	2. Pethuru Raj and Anupama C Raman, The Internet of Things – Enabling Technologies, Platforms, and use cases, CRC Press, Taylor and Francis, 2017 (https://www.vitalsource.com/en-uk/bookshelf/books/164705766).

Reference Books

1	Yasuura, H., Kyung, C.-M., Liu, Y., Lin, Y.-L., Smart Sensors at the IoT Frontier, Springer International Publishing.
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AUGMENTED REALITY AND VIRTUAL REALITY

Course Code	21AI644	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To develop the understanding of fundamentals and technological aspects of Augmented and Virtual Reality.
2	To learn about the basics of how to use Augmented Reality
3	To learn use of Augmented Reality in 3D Environment
4	Develop an understanding of Augmented Reality in Web development
5	To learn Virtual Reality and its applications

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understanding the fundamental concepts and principles of Augmented Reality (AR) and Virtual Reality (VR)	L2
CO2	Apply the basics of Augmented Reality using Image in different applications.	L3
CO3	Analyze and Apply the Augmented Reality on 3D Environment	L4
CO4	Analyze AR for Web application	L4
CO5	Understanding the Virtual reality and Human Interaction	L1

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4		3												
CO5														

Module	Topic	Hours	Assessment	CO
1	Module 1 : Introduction to Augmented Reality (AR) Basic Principles of AR and VR Technologies, Comparison between AR and Reapplications of AR and VR in various industries, Introduction of Tools and Software for AR and VR Development, Overview of interactive software development life cycle for AR and VR, Exploring Computer graphics using Unity Engine.	08	CIA, Quiz	C01
2	Module-2: Platform for AR , Introduction Different platform for AR: First Platform Vuforia database, setting up Vuforia. Image Target, Multiple Image Target, Adding Virtual Buttons, Practical examples and hands-	08	CIA, Quiz	CO2



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	on exercises to create different AR applications using images.			
3	Module-3: AR in 3D Environment Understanding Unity 3D: Installing package ARCore for Android or ARKit ,Learn how to detect and visualize surfaces in the real world to place virtual objects, Understand the importance of a placement indicator in AR applications , Explore the concept of ARRaycast and its role in placing objects accurately in the AR environment.	08	CIA,Quiz	CO3
4	Module-4: AR on Web: Introduction to WebGL, a JavaScript API for rendering interactive 2D and 3D graphics within any compatible web browser. Additionally, students will explore different frameworks for building AR experiences on the web, including Three.js and A-Frame.	08	CIA,Quiz	CO4
5	Module-5: Introduction to VR: Overview of Virtual Reality (VR), exploring its history, significance, and design principles. Students will learn about VR interfaces, controllers, and setting up VR hardware. Additionally, the module covers the XR plugin and toolkit for Unity, enabling students to create and test VR applications.	08	CIA,Quiz	CO5

Text Books	
1	G.C. Burdea & P. Coiffet, “Virtual reality Technology, Second Ed.”, Wiley-India, 2017
2	Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016
3	Augmented Reality: Principles & Practice by Schmalstieg / Hollerer, Pearson Education India; First edition (12 October 2016),ISBN-10: 9332578494
4	Augmented Reality _ Where We Will All Live, Jon Peddie

Reference Books	
1	D.A. Bowman et al., “3D User Interfaces: Theory and Practice”, Addison Wesley.
2	Creating Augmented and Virtual Realities: Theory and Practice for Next-Generation Spatial Computing by Erin Pangilinan , Steve Lukas, et al. 18 March 2019
3	Augmented Reality and Artificial Intelligence: The Fusion of Advanced Technologies (Springer Series on Cultural Computing) 30 April 2023 by Vladimir Geroimenko (Editor)



INTRODUCTION TO AI

Course Code	21AI651	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	OEC	Exam Hours	3

Course Objectives

1	To develop the understanding of fundamentals and technological aspects of Artificial Intelligence.
2	To learn about basics of how to use Artificial Intelligence.
3	To understand and explain the various Real World Applications of Artificial Intelligence.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Illustrate the ideas behind Artificial Intelligence, benefits, as well as current and future Challenges.	L2
CO2	Analyze and Apply the basics of Intelligent Agents, Examine the importance of formulating problems for solving Real world Applications.	L4
CO3	Understand and Discuss the basics Logical agents	L2
CO4	Discuss the concept of Knowledge representation	L1
CO5	Understand and Apply the basic ideas of sub domains of Artificial Intelligence.	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2		3												
CO3														
CO4														
CO5	3													

Module	Topic	Hours	Assessment	CO
1	Module-1: Introduction and History: What is AI? , Foundation of AI, History of AI, State of the Art in AI Intelligent Agents: Agents and Environments, Good Behavior: Concept of Rationality, Nature of Environments, Structure of Agents	08	CIA,Quiz	C01
2	Module-2: Solving Problem by Searching: Problem Solving Agents:Well-defined problems and solutions, Formulating problems, Example Problems: Toy problems, Real-world problems,Searching for solutions: Infrastructure for search algorithms,Measuring problem-solving performance, Uninformed Search Strategies- BFS, Uniform cost search DFS, Depth limited search, Iterative DFS, Bidirectional Search, Informed Search Strategies- Greedy best-first Search, A* search.	08	CIA,Quiz	C02



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3	Module-3: Logical Agents: Knowledge Based Agents, The Wumpus World, Logic, Propositional Logic: A Very Simple Logic, propositional Theorem Proving, Effective Propositional Model Checking, Agents Based on Propositional Logic	08	CIA, Quiz	C03
4	Module-4: Knowledge Representation: Ontological Engineering, Categories and Objects, Events, Mental Events and Mental Object	08	CIA, Quiz	C04
5	Module-5: AI Applications: Introduction to AI Domains: Data Science, Computer Vision, Natural Language Processing, AI Ethics: Moral Issues, Data Privacy, AI Bias, AI Access, Introduction to Robotics	08	CIA, Quiz	C05

Text Books	
1	Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J.
2	A First Course in Artificial Intelligence by Deepak Khemani

Reference Books	
1	NCERT Introduction to AI – Curate with the Support of Intel https://cbseacademic.nic.in/web_material/Curriculum21/publication/secondary/Class10_Facilitator_Handbook.pdf



NATURAL LANGUAGE PROCESSING LABORATORY

Course Code	21AIL66	Credits	1
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives

1	Explain the advantages and disadvantages of different NLP technologies and their applicability in different business situations.
2	Identify NLP applications in business and social contexts
3	Reflect on their own skill development
4	Organise and implement an NLP project in a business environment
5	Interpret the results of an NLP project.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply Python for text preprocessing, N-Gram modeling, POS tagging, and noun phrase extraction.	L4
CO2	Use ML for sentiment analysis, text summarization, and sentence generation.	L3
CO3	Implement NER and morphological analysis.	L4
CO4	Master language translation and named entity linking without NLP packages.	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3								2	2	2			2
CO2	3								2	2	2			2
CO3		3							2	2	2			2
CO4		3							2	2	2			2

Module	Topic	Hours	Assessment	CO
1	Study of Python and basic commands to access text data. (from notepad, pdf, word documents, online)	02	CIA, Quiz	CO1
2	A) Perform Pre-processing (Tokenization, stop word removal, unwanted character removal using Regex, stemming and lemmatization) of Text. B) Perform text pre - processing on a given corpus without using any pre - defined NLP packages. The text corpus given below: "The weather was beautiful. I went for a walk in the park. It was a sunny day, and the birds were chirping happily. Suddenly, a black cat crossed my path. I stopped and watched it disappear into the bushes. After that, I continued my stroll, enjoying the tranquility of nature."	02	CIA, Quiz	CO1



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3	A) Implement N-Gram model (Unigram, bigram, trigram extraction). B) Implement N -Gram model in python without using any predefined NLP packages. Note: use corpus of your own choice.	02	CIA,Quiz	CO1
4	A) Implement Part-of-Speech (POS) Tagging. B) Write a Python program to perform Part-of-Speech (POS) tagging on a given text corpus without using any predefined NLP packages. The text corpus and corresponding part-of-speech tags are provided below: "The cat chased the mouse around the house. Birds sang in the trees while the sun shone brightly in the sky. A group of children played happily in the park, laughing and running around."	02	CIA,Quiz	CO1
5	A) Implement chunking to extract Noun Phrases. B)Implement chunking to extract Noun and Verb phrases without using any pre defined NLP packages	02	CIA,Quiz	CO1
6	Sentence completion with words or phrases using random prompts.	02	CIA,Quiz	CO2
7	A)Using Machine Learning sentiment classification B)Implement machine learning sentiment classification without using any pre defined NLP packages: Training Corpus: ("I love this product", "positive"), ("This is excellent", "positive"), ("Terrible service", "negative"), ("It's okay, not great", "neutral"), ("Amazing experience", "positive"), ("Disappointing outcome", "negative"), ("Neutral feelings", "neutral"), ("I dislike it", "negative")	02	CIA,Quiz	CO2
8	A) Text summarization (Extractive and Abstractive) B) Implement Python programs for extractive and abstractive summarization without using predefined NLP libraries.	02	CIA,Quiz	CO2
9	A) Perform Name Entity Recognition (NER) on given text. B)Perform NER without using any pre-defined NLP packages The text corpus given below : "The capital of [France] is [Paris], a city known for its iconic [Eiffel Tower]. [John Smith] visited [Tokyo] last summer, exploring the bustling streets of [Shibuya Crossing]. [May 5th, 2023] marks the anniversary of a significant event in [history]. [Elon Musk] is the CEO of [SpaceX] and [Tesla]."	02	CIA,Quiz	CO3
10	A) Perform Morphological Analysis. B)Perform Morphological analysis without using any pre-defined NLP packages The text corpus given below : "The quick brown foxes jumped over the lazy dogs. Mary's cat is playing with a ball. Running swiftly, the athlete won the race. The painted houses lined the street, attracting curious onlookers."	02	CIA,Quiz	CO3
11	Perform dependency parsing without using any NLP packages	02	CIA,Quiz	CO4



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12	Perform language translation without NLP packages	02	CIA, Quiz	CO4
13	Named entity linking (NEL) without NLP packages	02	CIA, Quiz	CO4

Text Books	
1	Jurafsky, David, and James H. Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition. Upper Saddle River, NJ: Prentice-Hall, 2000. ISBN: 0130950696..
2	Manning, Christopher D., and Hinrich Schütze. Foundations of Statistical Natural Language Processing. Cambridge, MA: MIT Press, 1999. ISBN: 0262133601

Reference Books	
1	Frederick Jelinek, "Statistical Methods Of Speech Recognition", MIT Press, 1997
2	Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015
3	Nitin Indurkha and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
4	Richard M Reese, —Natural Language Processing with Java, O'Reilly Media, 2015
5	https://onlinecourses.nptel.ac.in/noc19_cs56/preview



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APPLIED BIGDATA AND CLOUD COMPUTING

Course Code	21AI62	Credits	01
L:T:P:S	0:0:2: 0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	3

Course Objectives	
1	To understand the need of Big Data, challenges and different analytical architectures
2	Installation and understanding of Hadoop Architecture and its eco systems
3	Processing of Big Data with Advanced architectures like Spark.
4	Describe graphs and streaming data in Spark

Course Outcomes		Blooms Taxonomy(highest)
CO1	Demonstrate a solid understanding of the principles, characteristics, and challenges of big data and cloud computing.	L4
CO2	Apply cloud computing concepts to effectively deploy, manage, and scale big data infrastructure and resources.	L3
CO3	Examine practical scenarios and real-life examples where the integration of Big Data analytics and cloud computing technologies has been applied to address various challenges.	L4
CO4	Develop analytical skills to systematically analyze and select optimal tools based on problem analysis.	L4
CO5	Implementing, and deploying a big data analytics solution on a cloud computing platform	L4
CO6	Create simple software programs and web applications, as well as effectively utilize programming tools and environments	L5

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3					3									
CO4		3												
CO5		3												
CO6					3				2	2				

Module	Topic	Hours	Assessment	CO
1	Install Virtual box/VMware Workstation with different flavors of Linux or windows OS on topofwindows7or8.	02	CIA,Quiz	CO2



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2	Install a C compiler in the virtual machine created using virtual box Write a C program for storing data in a simulated cloud storage environment using a local file	02	CIA, Quiz	C01 C06
3	Write a C-Program for CPU usage Monitoring and Logging on Cloud-Based-Ubuntu Servers.	02	CIA, Quiz	C06
4	Use Google App Engine Launcher to launch the web applications	02	CIA, Quiz	C04 C06
5	Installation of Single Node Hadoop Cluster on Ubuntu 22.04 LTS.	02	CIA, Quiz	C04 C05
6	Hadoop Programming: Word Count Map Reduce Program Using Eclipse	02	CIA, Quiz	C06
7	File Management tasks in Hadoop using HDFS commands a. Adding files to HDFS b. Retrieving files from HDFS c. Deleting files from HDFS	02	CIA, Quiz	C05 C06
8	Creation of Data Frames using Create Data Frame working with an example related to FIFA dataset.	02	CIA, Quiz	C05
9	Simulator program: Analyze the average, minimum and maximum response time and datacenter Processing time and write the findings.	02	CIA, Quiz	C04
10	Conduct experimentation by changing different service broker policies to analyze the average, minimum and maximum response time and total datacenter Processing time. Also, analyze the total cost required to run the application.	02	CIA, Quiz	C04

Text Books	
1	Boris lublinsky, Kevin t. Smith, Alexey Yakubovich, Professional Hadoop Solutions, Wiley, 2015.
2	Tom White, Hadoop: The Definitive Guide, OReilly Media Inc., 2015
3	Garry Turkington, Hadoop Beginners Guide, Packt Publishing, 2013.
4	Eric Sammer, "Hadoop Operations", 1st Edition, OReilly, 2012.

Reference Books	
1	Pethuru Raj, Anupama Raman, DhivyaNagaraj and Siddhartha Duggirala, HighPerformance Big-Data Analytics: Computing Systems and Approaches, Springer, 2015.
2	Jonathan R. Owens, Jon Lentz and Brian Femiano, Hadoop Real-World Solutions Cookbook, Packt Publishing, 2013
3	https://onlinecourses.nptel.ac.in/noc20_cs92/preview



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MANAGEMENT AND ENTREPRENEURSHIP
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Course Code	21HS71	Credits	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	HSMC	Exam Hours	3

Course Outcomes: After completion of the course, the graduates will be able to

MANAGEMENT & ENTREPRENEURSHIP	
CO1	Apply the principles of management in business activities.
CO2	Use the managerial and entrepreneurial qualities & skills under real world condition.
CO3	Analyze the functions of Management & Entrepreneurship and apply those in practical situations.
CO4	Identify various schemes provided by government of India to support business enterprise.
CO5	Develop leadership skills to build a small scale industry.
CO6	Develop entrepreneurial personality, able to prepare project report and initiate SSI.

Mapping of Course outcomes to Program outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	2		3	2	-	-	2
CO2	-	-	-	-	-	2		3	2	-	-	2
CO3	-	-	-	-	-	2		3	2	-	-	2
CO4	-	-	-	-	-	2		3	2	-	2	2
CO5	-	-	-	-	-	2		3	2	-	2	2
CO6	-	-	-	-	-	2		3	2	-	2	2

Unit	Course Content	Hours	COs
1	MANAGEMENT: Introduction – Meaning – nature and characteristics of Management, Scope and Functional areas of management – Management as a science, art of profession – Management & Administration – Roles of Management, Levels of Management. PLANNING: Nature, importance and purpose of planning process – Objectives – Types of plans (Meaning Only) – Decision making –Importance of planning. (Case studies discussion)	08	CO1 CO2
2	ORGANIZING AND STAFFING: Nature and purpose of organization – Principles of organization – types of organization – Departmentation – Committees- Centralization Vs Decentralization of authority and responsibility – Span of control, Nature and importance of staffing. (Case studies discussion)	08	CO1 CO2
3	DIRECTING & CONTROLLING: Meaning and nature of directing – Leadership styles, Maslow’s theory of motivation, Communication – Meaning and importance – coordination, meaning and importance and Techniques of Co-Ordination. Meaning and steps in controlling — Methods of establishing control (in brief). (case studies discussion)	08	CO3 CO4
4	ENTREPRENEUR: Meaning of Entrepreneur; Evolution of the Concept, Functions of an Entrepreneur, Types of Entrepreneur, Entrepreneur – an emerging Class. Concept of Entrepreneurship – Evolution of Entrepreneurship, Development of Entrepreneurship; Stages in entrepreneurial process; Role of entrepreneurs in Economic Development; Entrepreneurship – its Barriers. (Case studies discussion, role play / group discussion)	08	CO3 CO4
5	SMALL SCALE INDUSTRY: Definition; Characteristics; Need and rationale: Objectives; Scope; role of SSI in Economic Development. Advantages of SSI, Steps to start an SSI, Impact of Liberalization, Privatization, Globalization on S.S.I, Effect of WTO/GATT. Startup India: Benefits, Policies. Action plan- simplification and Handholding, Funding Support and incentives, Industry-Academia Partnership and Incubation. Strategies encouraging entrepreneurship through NAIN. Venture capitalist, SSI funding schemes by banks and financial institutions, Government of India Initiatives on Thrust Areas, overview of detailed project report/profile. (Related case studies, supporting videos and group discussion)	08	CO5 CO6

Text Books	
1	Principles of Management – P.C.Tripathi, P.N.Reddy – Tata McGraw Hill.
2	Dynamics of Entrepreneurial Development & Management – Vasant Desai – Himalaya Publishing House.
3	Entrepreneurship Development – Poornima.M.Charantimath – Small Business Enterprises – Pearson Education – 2006 (2 & 4).
4	Management & Entrepreneurship-N V R Naidu, IK International, 2008

Reference Books	
1	Management Fundamentals – Concepts, Application, Skill Development – Robers Lusier – Thomson.
2	Entrepreneurship Development – S.S.Khanka – S.Chand & Co.
3	Management – Stephen Robbins – Pearson Education/PHI – 17th Edition, 2003.
4	http://www.startupindia.gov.in/
5	http://startup.karnataka.gov.in/docs/Startup_Policy_Karnataka.pdf



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GENERATIVE AI

Course Code	21AI72	Credits	3
L:T:P:S	2:2:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives	
1	Apply text mining techniques
2	Analyze text with advanced models
3	Implement AI architectures
4	Analyze the working of GEN AI models

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply unstructured data analysis	L3
CO2	Apply advanced models in unstructured data analysis	L3
CO3	Analyze the GEN AI architecture	L4
CO4	Analyze the working of GEN AI models	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3				2									3
CO2	3				2									3
CO3		3												3
CO4		3												3

Module	Topic	Hours	Assessment	CO
1	Introduction to text mining, Text Preprocessing, N-grams, Text Normalization – Tokenization, Stemming and Lemmatization, Regex Basics, Bag of word Models, Text Vectorization, DTM and TDM, Text Corpus and Wordcloud, Dimensionality reduction, Singular Value Decomposition, Handling Sparsity, Text multiclass classification, Text clustering with K-means and DBSCAN, Embeddings, TF, TF-IDF, Word2Vec, RNN	8	CIA, SEE, Quiz	1,2
2	Attention mechanism: Why attention, Self-attention, Encoders and Decoders, Variational encoders, Introduction to BERT, BERT Architecture, Boltzmann machine network, Generative Modelling: Generative Modelling and AI, Generative Modelling Framework	8	CIA, SEE, Quiz	2,3,4



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3	Representation learning, GANs (Generative Adversarial Network), Autoregressive Models: LSTM. RNN Extensions: Stacked RNN, Gated RNN, Bidirectional RNN. Diffusion Models, Transformers: GPT,T5, GPT 3, GPT-4, 8Controllers, training Controllers	8	CIA,SEE, Quiz	3,4
4	Large Language Models, Transfer Learning,Introduction to Energy based models, Content Generation – music Pixel CNN: Masked Convolution, Mixture Distribution	8	CIA,SEE, Quiz	3,4
5	Reinforcement Learning, Prompt Engineering, Multimodal Models, DALLE architecture, LLAMMAarchitecture, The future of Gen AI, Gen AI Ethics and Challenges	8	CIA,SEE, Quiz	3,4

Text Books	
1	https://web.stanford.edu/~jurafsky/slp3/ed3book_jan72023.pdf
2	Mathew Windham, Unstructured Data Analysis, Wiley Publications, 2018
3	Joseph Babcock and Raghav Bali, Generative AI with Python and TensorFlow, O'Reilly Publications, 2021
4	Forsyth and Ponce, Computer Vision: A Modern Approach,, Pearson, 2nd Edition

Reference Books	
1	David Foster, Generative Deep Learning, O'Reilly Media, Inc, 2023
2	For the second part we will mostly follow "Multiple View Geometry in Computer Vision", by Hartley and Zisserman, CambridgeUniversity Press



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ADVANCED DATA PRIVACY AND SECURITY

Course Code	21AI731	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Understand fundamental principles of Cyber security and Data privacy for protecting data in given use cases.
2	Apply the data protection and privacy-preserving technique for given scenarios.
3	Analyse the effective countermeasures to protect against data threats.
4	Investigate the various tools available in literature and write concise experiment reports that detail the methods, results, and conclusions of tools.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand fundamental principles of Cyber security and Data privacy for protecting data in given use cases.	L2
CO2	Apply the data protection and privacy-preserving technique for given scenarios..	L3
CO3	Analyse the effective countermeasures to protect against data threats.	L4
CO4	Investigate the various tools available in literature and write concise experiment reports that detail the methods, results, and conclusions of tools.	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	Introduction to Cybersecurity, Confidentiality, Integrity, and Availability. Foundations of Privacy - Information privacy, Measurement, Theories. Data Types, States, PII, introduction to Data Privacy and its importance. Data breaches and case studies	8	CIA,SEE, Quiz	1,3
2	Phishing and Identity Theft: Introduction, Methods of Phishing, Phishing Techniques, Spear Phishing, Types of Phishing Scams, Phishing Toolkits and Spy Phishing, Phishing Countermeasures, Identity Theft (ID Theft). Types of Malware, Data exfiltration, Ransomware case study with multi-level extortion.	8	CIA,SEE, Quiz	2,3,4



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3	History of Cryptography, Cryptographic techniques, Cryptographic algorithms. Symmetric and Asymmetric Encryption, Applications of Encryption with case studies. Hash functions, Obfuscation techniques and Digital signatures Authentication and Authorization	8	CIA,SEE, Quiz	2,3,4
4	Privacy-preserving mechanisms Privacy Models / Anonymization models: K-anonymity, l-diversity, t-closeness, differential privacy, implementing anonymization. Using technology for preserving privacy. Statistical Database security, Inference Control, Privacy-preserving Data mining, Hippocratic databases Social Network Privacy, Location Privacy, Query Log Privacy and Biomedical Privacy.	8	CIA,SEE, Quiz	2,3,4
5	Security management, Governance Risk and compliance (GRC), GRC framework, security standards. Contingency planning - Incidence response, Disaster Recovery, BCP. Privacy Regulation: Data Protection, DPDP Act 2023, GDPR, PCI-DSS, HIPAA. DPO responsibilities and Incidence reporting (case study with sample report).	8	CIA,SEE, Quiz	2,3,4

Text Books

1	"Phishing and Countermeasures: Understanding the Increasing Problem of Electronic Identity Theft" by Markus Jakobsson and Steven Myers.
2	William Stallings , Cryptography and Network Security Principles And Practice Pearson Education Fourth Edition.
3	"Privacy-Preserving Data Publishing: Concepts and Techniques" by Benjamin C.M. Fung, Ke Wang, Rui Chen, and Philip S. Yu
4	"Data Privacy Law: A Practical Guide to Compliance" by Gwen Kennedy

Reference Books

1	Introduction to Privacy-Preserving Data Publishing: Concepts,Benjamin C.M. Fung, Ke Wang, Ada Wai-Chee Fu -2010
2	Alfred J. Menezes, Paul C. vanorschot and Scott A. Vanstone, "Handbook of Applied Cryptography" CRC Press.
3	Current and Emerging Trends in Cyber Operations: Policy, Strategy and Practice (Palgrave Macmillan's Studies in Cybercrime and Cybersecurity) 1st Edition. 2015 Edition



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SYSTEM ARCHITECTURE AND DESIGN

Course Code	21AI732	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Apply core design principles
2	Analyze system design components
3	Investigate scaling and database solutions
4	Investigate real-world system design

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply core design principles	L3
CO2	Analyze system design components	L4
CO3	Investigate scaling and database solutions	L4
CO4	Investigate real-world system design	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3													
CO2		3												
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	Introduction: basic Low level design, Constructors, UML designs and types: UML & Object diagram, activity diagram, sequence diagram, state diagram	8	CIA,SEE, Quiz	1
2	Design Principles: Learning Core design principle, SOLID, GRASP, DRY, KISS, Advance LLD and usecases: Single responsibility & open/closed principle, Liskov substitution & interface segregation principle, dependency inversion	8	CIA,SEE, Quiz	1
3	System Design Components: Domain name system: Introduction to DNS, request routing, DNS Cache, DNS in action :Route 53 Load Balancer: Types of load balancer, scaling load balancer, load balancing algorithms	8	CIA,SEE, Quiz	2
4	Scaling and Database, Horizontal scaling, vertical scaling Database Architecture: Relational database, non relational database, database replication, database sharding, advance high level design, caching, System design Framework	8	CIA,SEE, Quiz	3



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5	Case Studies: Movie Ticket Booking System, Airline Booking System, Netflix system design	8	CIA,SEE, Quiz	4
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Text Books	
1	Design Patterns: Elements of Reusable Object-Oriented Software” by Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides
2	“Software Engineering”, International Computer Science Series, Ian Somerville, Pearson Education, 10th Edition, 2020.



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HIGH PERFORMANCE COMPUTING

Course Code	21AI733	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Apply HPC concepts and architectures
2	Analyze Parallel computing models and metrics
3	Investigate Parallel algorithms and programming techniques
4	Investigate optimization of HPC system

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply HPC concepts and architectures	L3
CO2	Analyze Parallel computing models and metrics	L4
CO3	Investigate Parallel algorithms and programming techniques	L4
CO4	Investigate optimization of HPC system	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3													
CO2		3												
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	Introduction to High Performance Computing: Introduction to HPC- Definition and significance, Historical context and evolution, Applications of HPC Computer Architecture for HPC- Modern CPU architectures, Memory hierarchy and impact on performance, Parallel computing architectures: SMP, MPP, Cluster, Grid, Cloud Computing, GPGPU (General-Purpose computing on Graphics Processing Units) Performance Metrics and Benchmarking : Metrics: FLOPS, throughput, latency, speedup, efficiency, scalability, Amdahl's Law and Gustafson's Law, Benchmarking tools and suites (LINPACK, SPEC, NAS Parallel Benchmarks)	8	CIA, SEE, Quiz	1
2	Parallel Computing Models and Paradigms: Parallel Computing Models- Shared memory vs. distributed memory systems, Data parallelism vs. task parallelism, SIMD (Single Instruction, Multiple Data) and MIMD (Multiple Instruction, Multiple Data) architectures	8	CIA, SEE, Quiz	2,3



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	Parallel Programming Models-MPI (Message Passing Interface), OpenMP (Open Multi-Processing), CUDA (Compute Unified Device Architecture), OpenCL (Open Computing Language)			
3	Parallel Algorithms and Programming: Parallel Algorithm Design-Basics of parallel algorithm design, Decomposition techniques: data decomposition, task decomposition, Parallel algorithms for sorting, searching, matrix operations, graph algorithms Programming for HPC-Introduction to parallel programming languages and libraries, Practical examples using MPI, Shared memory programming with OpenMP, GPU programming with CUDA and OpenCL	8	CIA,SEE, Quiz	2,3
4	Performance Optimization and Tuning: Performance Analysis and Optimization- Profiling and performance analysis tools, Compiler optimizations and code tuning, Memory optimization techniques Load Balancing and Dynamic Scheduling- Load balancing strategies, Dynamic scheduling techniques High Performance Libraries and Frameworks- Numerical libraries (BLAS, LAPACK, ScaLAPACK), HPC frameworks (Hadoop, Spark, TensorFlow)	8	CIA,SEE, Quiz	4
5	Case Studies, Applications, and Future Trends: Case Studies and Applications- Scientific computing applications (e.g., climate modeling, molecular dynamics), Industrial applications (e.g., financial modeling, oil and gas exploration), HPC in AI and machine learning Future Trends in HPC-Exascale computing, Quantum computing, Neuromorphic computing, Trends in HPC hardware and software	8	CIA,SEE, Quiz	2,3,4

Text Books

1	Introduction to High Performance Computing for Scientists and Engineers" by Georg Hager and Gerhard Wellein
2	"Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers" by Barry Wilkinson and Michael Allen
3	"High Performance Computing: Modern Systems and Practices" by Thomas Sterling, Matthew Anderson, and Maciej Brodowicz

Reference Books

1	"Using OpenMP: Portable Shared Memory Parallel Programming" by Barbara Chapman, Gabriele Jost, and Ruud van der Pas
2	"Programming Massively Parallel Processors: A Hands-on Approach" by David B. Kirk and Wen-mei W. Hwu
3	"Parallel Programming with MPI" by Peter Pacheco
4	"High Performance Computing: Paradigm and Infrastructure" edited by Laurence T. Yang and Minyi Guo



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QUANTUM COMPUTING

Course Code	21AI734	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To understand the fundamentals and technological aspects of Quantum Computing.
2	To learn the basics of how Quantum Computing Models & Algorithms work.
3	To understand and explain Quantum Operations.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand the foundational principles of quantum computation and the Quantum Circuit Model	L2
CO2	Apply mathematical tools & techniques in context of quantum mechanics and computation.	L3
CO3	Analyze and critique quantum algorithms and their efficiency.	L4
CO4	Investigate quantum error correction codes for fault-tolerant quantum computation.	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	INTRODUCTION AND BACKGROUND: Overview, Computers and the Strong Church–Turing Thesis, The Circuit Model of Computation, A Linear Algebra Formulation of the Circuit Model, Reversible Computation LINEAR ALGEBRA AND THE DIRAC NOTATION: The Dirac Notation and Hilbert Spaces, Dual Vectors, Operators, The Spectral Theorem, Functions of Operators, Tensor Products, The Schmidt Decomposition Theorem	8	CIE,SEE, Quiz	1
2	QUBITS AND THE FRAMEWORK OF QUANTUM MECHANICS: The State of a Quantum System, Time-Evolution of a Closed System, Composite Systems, Measurement, Mixed States and General Quantum Operations, Mixed States, Partial Trace, General Quantum Operations.	8	CIE,SEE, Quiz	2
3	A QUANTUM MODEL OF COMPUTATION: The Quantum Circuit Model, Quantum Gates, 1-Qubit Gates, Controlled-U Gates, Universal Sets of Quantum Gates, Efficiency of Approximating Unitary Transformations, Implementing Measurements with Quantum Circuits,	8	CIE,SEE, Quiz	1



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	SUPERDENSE CODING AND QUANTUM TELEPORTATION: Superdense Coding, Quantum Teleportation, An Application of Quantum Teleportation.			
4	QUANTUM ALGORITHMS: Probabilistic Versus Quantum Algorithms, Phase Kick-Back, The Deutsch Algorithm, The Deutsch-Jozsa Algorithm, Simon's Algorithm, Grover's Quantum Search Algorithm.	8	CIE,SEE, Quiz	3
5	QUANTUM ERROR CORRECTION: Classical Error Correction, The Error Model, Encoding, Error Recovery, The Classical Three-Bit Code, Fault Tolerance, Quantum Error Correction, Error Models for Quantum Computing, Encoding, Error Recovery, Three- and Nine-Qubit Quantum Codes, The Three-Qubit Code for Bit-Flip Errors, The Three-Qubit Code for Phase-Flip Errors, Quantum Error Correction Without Decoding, The Nine-Qubit Shor Code, Fault-Tolerant Quantum Computation, Concatenation of Codes and the Threshold Theorem	8	CIE,SEE, Quiz	4

Text Books	
1	Phillip Kaye, Raymond Laflamme et. al., An Introduction to Quantum Computing, Oxford University press, 2007
2	Chris Bernhardt, Quantum Computing for Everyone, The MIT Press, Cambridge, 2020

Reference Books	
1	Quantum Computation and Quantum Information, M. A. Nielsen & I. Chuang
2	Cambridge University Press (2013). 2. Quantum Computing, A Gentle Introduction, Eleanor G. Rieffel and Wolfgang H. Polak MIT press (2014)



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BUSINESS INTELLIGENCE

Course Code	21AI741	Credits	2
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To understand the fundamentals of Business Intelligence.
2	To understand how BI supports in better decision-making and strategic planning within an organization.
3	To learn about business intelligence methods & to apply in various situations

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand the foundational concepts and frameworks of Business Intelligence (BI) and Decision Support Systems (DSS)	L2
CO2	Apply decision-making models and techniques supported by BI and DSS	L3
CO3	Evaluate the impact of AI on Business Intelligence and decision-making processes	L5
CO4	Design strategies for implementing and managing AI-powered BI solutions	L6

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3													
CO3		3												
CO4			3											

Module	Topic	Hours	Assessment	CO
1	Business Intelligence Data Analytics and Decision Support: Opening Vignette, Changing Business Environment and Computerized Decision Support, Managerial Decision making, Information System Support for Decision Making, An Early Framework for Computerized Decision Support, The Concept of Decision Support Systems (DSS), A Framework for Business Intelligence, The Work System View of Decision Support, Overview of Business Analytics, Introduction to Big Data Analytics	8	CIA,SEE, Quiz	1
2	Decision Making Technologies: Decision Making, Models, Phases of Decision Making Process, The intelligent Phase, The Design Phase , The Choice Phase, The implementation phase, How Decisions are supported, DSS- Capabilities, classifications and components AI and AI Powered Analytics: AI and its Rise in Modern Enterprises, How AI changes the Enterprise BI Landscape, The Analytics Sphere, BI enabled vs AI Powered Analytics, From Data to intelligent Decision Making to Insightful Decisions, AI powered Techniques for BI	8	CIA,SEE, Quiz	2



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3	Data Warehousing: Data Warehousing Definitions and Concepts, Data Warehousing Process Overview, Data warehousing Architecture, Data Integration and the extraction, transformation and load, Processes, Data Warehouse Development, Implementation issues, Administration, Security issues and Future Trends.	8	CIA,SEE, Quiz	2
4	Knowledge Management and Collaborative Systems: Introduction to Knowledge Management, Organizational Learning and Transformation, Knowledge Management Activities, Approaches to Knowledge Management, Information Technology (IT) In Knowledge Management, Knowledge Management Systems Implementation, Making Decisions in Groups, Supporting Group work with Computerized Systems, Tools for indirect support of Decision Making, Direct Computerized Support for Decision Making	8	CIA,SEE, Quiz	3
5	Business Analytics For Emerging Trends and Future Impact: Location based Analytics for Organizations, Analytics and Applications for Consumers, Recommendation Engines, Web 2.0 and Online Social Networking, Cloud Computing and BI. What's Next when AI Meets BI: AI powered Cognitive Computing, Security and Governance in AI Powered BI, The Trust Factor in AI powered BI	8	CIA,SEE, Quiz	4

Text Books	
1	Business Intelligence, A Managerial Perspective on Analytics, Sharda R, Delen D, Turban E, 10 th Edition, Pearson, 2015
2	AI meets BI Artificial Intelligence and Business Intelligence, Lakshman Bulusu, Rosendo Abellera, CRC Press , 2021

Reference Books	
1	Larissa T. Moss, S. Atre, —Business Intelligence Roadmap: The Complete Project Lifecycle of Decision Making, Addison Wesley, 2003.
2	Carlo Vercellis, —Business Intelligence: Data Mining and Optimization for Decision Making, Wiley Publications, 2009.
3	David Loshin Morgan, Kaufman, —Business Intelligence: The Savvy Manager's Guidel, Second Edition, 2012.
4	Cindi Howson, —Successful Business Intelligence: Secrets to Making BI a Killer Appl, McGraw- Hill, 2007



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KNOWLEDGE REPRESENTATION

Course Code	21AI742	Credits	2
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Apply Knowledge representation techniques
2	Analyze reasoning methods and algorithms
3	Analyze rule based and object based representations
4	Analyze inheritance and default reasoning strategies

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply Knowledge representation techniques	L3
CO2	Analyze reasoning methods and algorithms	L4
CO3	Analyze rule based and object based representations	L4
CO4	Analyze inheritance and default reasoning strategies	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3													
CO2		3												
CO3		3												
CO4		3												

Module	Topic	Hours	Assessment	CO
1	MODULE 1 The Key Concepts: Knowledge, Representation, and Reasoning, Why Knowledge Representation and Reasoning? The Role of Logic. The Language of First-Order Logic: Introduction, The Syntax, The Semantics, The Pragmatics, Explicit and Implicit Belief. Expressing Knowledge: Knowledge Engineering, Vocabulary, Basic Facts, Complex Facts, Terminological Facts, Entailments, Abstract Individuals, and Other Sorts of Facts.	8	CIE,SEE, Quiz	1
2	UNIT II Resolution: The Propositional Case, Handling Variables and Quantifiers, Dealing with Computational Intractability. Reasoning with Horn Clauses: Horn Clauses, SLD Resolution, Computing SLD Derivations.	8	CIE,SEE, Quiz	2,3



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3	UNIT III Procedural Control of Reasoning: Facts and Rules, Rule Formation and Search Strategy, Algorithm Design, Specifying Goal Order, Committing to Proof, Controlling Backtracking, Negation as, And Dynamic Databases. Rules in Production Systems: Production Systems: Basic Operation, Working Memory, Production Rules, A First Example, A Second Example, Conflict Resolution, Making Production Systems More Efficient, Applications and Advantages, Some Significant Production Rule Systems .	8	CIE,SEE, Quiz	2,3
4	UNIT IV Object-Oriented Representation: Objects and Frames, A Basic Frame Formalism, An Example: Using Frames to Plan a Trip, Beyond the Basics. Structured Descriptions: Descriptions, Meaning and Entailment, Computing Entailments, Taxonomies and Classification, Beyond the Basics.	8	CIE,SEE, Quiz	3,4
5	UNIT V Inheritance: Inheritance Networks, Strategies for Defeasible Inheritance, The Shortest Path Heuristic, Problems with Shortest Path, Inferential Distance, A Formal Account of Inheritance Networks. Defaults: Introduction, Closed-World Reasoning, Circumscription, Default Logic, Auto epistemic Logic.	8	CIE,SEE, Quiz	3,4

Text Books	
1	Ronald J. Brachman, Hector J. Levesque with a contribution by Maurice Pagnucco “KNOWLEDGE REPRESENTATION AND REASONING” Morgan Kaufmann Publishers is an Imprint of Elsevier., 2004 by Elsevier, Inc. All rights reserved.
2	Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell



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SOCIAL NETWORK ANALYSIS

Course Code	21AI743	Credits	2
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Understand the components and entities of the social network
2	Analyze social network data to comprehend user relationships and recommend the essential information appropriate
3	Model and visualize the social network
4	Understand the components and entities of the social network

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply the basic concepts of social networks and identify prominence of elements	L3
CO2	Analyze social networks using statistical methods	L4
CO3	Analyze the structure and patterns of networks	L4
CO4	Examine various social network models and their properties.	L4

CO-PO-PSO Mapping														
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3													
CO2		3			2									
CO3		3			2									
CO4		3			2									

Module	Topic	Hours	Assessment	CO
1	Social Network Perspective , Fundamentals concepts in Network Analysis: Sociogram, Sociometry. Social Network Data: Types of Networks: One-Mode, Two-Mode, Affiliation, Ego-centered and Special Dyadic Networks, Network Data, Measurement and Collection Notations for Social Network Data : Graphs, Directed, Singed, Valued graphs, Multigraph, Relations and Matrices.	8	CIA,SEE, Quiz	1
2	Centrality and Prestige : Prominence: Actor-Centrality, Prestige, Group-Centrality, Prestige, Non directional Relations-Degree, Closeness, Betweenness, Eigen Vector Centrality, Directional Relations-Centrality, Prestige. Structural Balance and Transitivity : Structural Balance: Signed Non directional, Signed Directional Relations, Checking for Balance, Index for Balance, Clusterability-Theorems, Clustering Coefficient and Transitivity	8	CIA,SEE, Quiz	1,3



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3	Cohesive Subgroups: Social Group and Subgroup-Notation, Subgroups Based on Complete Mutuality: Clique, Reachability and Diameter: n-cliques, n-clans and n-clubs, Subgroups Based on Nodal Degree: k-plexes, k-cores, Community detection, Link Prediction Roles and Positions: Structural Equivalence: Definition, Social Roles and , Positional Analysis, Measuring Structural Equivalence, Representation of Network Positions	8	CIA,SEE, Quiz	3
4	Dyadic and Triadic Methods: Dyads: Definitions, Dyad Census, Index, Simple Distributions, Triads: Random Models and Substantive Hypotheses, Triad Census, Distribution of a Triad Census- Mean and Variance, Testing Structural Hypotheses	8	CIA,SEE, Quiz	2
5	Models in Social Network: Small world network, Watt Strogatz networks, statistical models for social networks, network evaluation model, Preferential attachment, power law, Random Model : Erdos - Renyi model, Barabasi Albert model, Epidemic model, Case study: Text and opinion Analysis	8	CIA,SEE, Quiz	4

Text Books	
1	Wasserman Stanley, and Katherine Faust, Social Network Analysis: Methods and Applications, Structural Analysis in the Social Sciences. Cambridge University Press, 2012 Online Edition.
2	Albert-László Barabási, Network Science, Cambridge University Press, 1st edition, 2016.
3	Vito Latora, Vincenzo Nicosia, Giovanni Russo, Complex Networks Principles, Methods and Application, Cambridge University Press,2017.

Reference Books	
1	Tanmoy Chakraborty, Social Network Analysis, Wiley,2021



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APPLIED AI

Course Code	21AI744	Credits	2
L:T:P:S	2:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Understand how to use AI for solving complex problems
2	Understand the ethics in handling sensitive data
3	Apply AI to solve real world problems
4	Analyze how modern AI practices are done in various industry

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply AI techniques in various domain	L3
CO2	Apply ethics of AI while working in appropriate domain	L3
CO3	Analyze where to use AI techniques in a given scenario	L4
CO4	Analyze how AI is used to solve complex real-world problems	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3								2	1			N/A		
CO2	3								2	1					
CO3		3													
CO4		3													

Module	Topic	Hours	Assessment	CO
1	AI in Healthcare: Using AI for Disease Diagnosis and Patient Monitoring, Natural Language Processing and Data Analytics in Health Care, Patient Risk Stratification and Augmenting Clinical Workflows Taking, an Integrated Approach to Hospital Management and Optimization	5	CIA,SEE, Quiz	1,3,4
2	AI in Fintech: AI & ML in Finance, Open API & Marketing Channels, Blockchain Technology & Cryptocurrencies, AI in Payments, Credit & Lending, Challenge for AI in, Banks Trading & Capital Markets, Insurance	5	CIA,SEE, Quiz	1,3,4
3	AI in Manufacturing: Application of expert systems for the economic evaluation of machining operations. Integrating enterprise automation with an intelligent agent framework. Introduction to digital twin technology and cyber-physical manufacturing systems. Case Study in Robotic Service	5	CIA,SEE, Quiz	1,3,4
4	AI in Design and Gaming: AI Design Tools in Fashion and Graphic Design, ML agents in Gaming, Reinforcement Learning, A* algorithm in gaming. Unity Tool for ML agents for self-car parking simulation, A* for path finding.	5	CIA,SEE, Quiz	1,3,4



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5	Ethics in AI: Principles of ethics in AI, exploring the moral implications and societal impacts of AI technologies., Guidelines in AI Ethics, AI in Practice, Advances in AI Ethics emerging issues, and innovative approaches to addressing ethical concerns in AI. Applying ethics in Health care, Finance etc.	6	CIA,SEE, Quiz	2
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Text Books	
1	"Ethics of Artificial Intelligence and Robotics" by Vincent C. Müller
2	LLM and Generative AI for Healthcare: The Next Frontier by Kerrie Holley (Author), Manish Mathur (Author) Oreilley publication
3	AI-First Healthcare: AI Applications in the Business and Clinical Management of Health by Kerrie L. Holley (Author), Siupo Becker M. D. (Author) Oreilley publication
4	A. Fazel Famili (Editor), Dana S. Nau (Editor), Steven H. Kim (Editor); Artificial Intelligence Applications in Manufacturing, AAAI Press.
5	Ellen Friedman, Ted Dunning, AI and Analytics in Production; O'Reilly Media, Inc., 2018 (ISBN: 9781492044116)
6	AI for Game Developers: Creating Intelligent Behavior in Games by David M Bourg (Author), Glenn Seemann (Author) Oreilly Publication

Reference Books	
1	Ethical Machines by Reid Blackman published by Harvard Business Review Press 2022
2	AI for Finance By Edward K. Tsang published by CRC Press June 2023
3	Designing Autonomous AI by Kence Anderson published by O'Reilly Media, Inc. June 2022
4	Generative AI in Practice by Bernard Merr, Wiley,2024
5	Ethical Machines by Reid Blackman published by Harvard Business Review Press 2022

Links	
1	https://unity.com/blog/engine-platform/training-intelligent-adversaries-using-self-play-with-ml-agents
2	https://ai.sony/projects/gaming_ai/
3	https://federated.withgoogle.com/



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AI IN PRACTICE

Course Code	21AI751	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	OEC	Exam Hours	3

Course Objectives

1	To Understand the foundational concepts of Artificial Intelligence & related Fields.
2	Gain insights in identifying and collecting data from various sources and their effective preprocessing.
3	Explore the transformative impact of AI across multiple sectors.
4	Explore advanced AI concepts such as deep learning along with integration of AI with IoT.

Course Outcomes

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understanding the fundamentals of AI and its associated domains	L2
CO2	Apply the knowledge of AI technologies in real world scenarios.	L3
CO3	Analyzing the applications of AI across multiple industries.	L4
CO4	Examine AI solutions in Domains like Life sciences, Social sciences, Engineering sciences and other emerging technologies	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1															
CO2		3												2	
CO3		3												2	
CO4	3								2	2		2		2	

Module	Topic	Hours	Assessment	CO
1	Artificial Intelligence Introduction, Types of AI, Introduction to Machine learning, types of machine learning. Common applications of AI Trailblazers: Using Artificial Intelligence to Power The Retail And Business-To-Business Services of The Future, Maximizing The Potential Of Artificial Intelligence, Using Deep Learning To Drive Business Performance, Cognitive Computing, Artificial Intelligence as a Part Of The Fabric Of Everyday Life, Automating Retail With Artificial Intelligence.	8	CIA+SEE+Quiz	1,3,4
2	Data Collection & Cleaning: Data Sources, Methods of Data Collection, Handling Missing Values. Feature scaling and normalization, Feature selection techniques. Case Studies on Retail, Consumer Goods and Food and Beverage: Leverage AI to enhance luxury sales, Maintain market leadership, Analyze customer data & automate processes, Improve efficiency in food service, AI and robotics for home and workplace, Fashion retail innovation, Recruitment, Inventory management.	8	CIA+SEE+Quiz	3,4

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3	Model Selection: How to choose the right model? Training, Test & Validation processes, Model Performance Metrics, Confusion matrix. Case studies on Media, Entertainment and Telecom Companies: Creating memorable interactions, Improving user experiences in TV viewing, Combat online bullying, Address the skills crisis, Connecting Communities with AI solutions, Fake News And Spambots, Improve service quality assessment, Video streaming.	8	CIA+SEE+Quiz	3,4
4	Introduction to deep learning: Introduction to Neural Networks, Activation Functions, Loss functions, Architecture of CNN & RNN, Applications of Neural Networks. Case studies on Services, Financial and Healthcare Companies: Fraud Detection, Medical decisions & Research, Health analysis, Combat Counterfeit Industries, Simplify mortgage processes, Online Transactions, Understand customers.	8	CIA+SEE+Quiz	3,4
5	Introduction to IoT Integration in AI: Components of IoT, How IoT and AI Complement Each Other, Microcontrollers and Development Boards (Arduino, Raspberry Pi). Introduction to generative AI. Case studies on Manufacturing, Automotive, Aerospace and Industry 4.0 Companies: Develop and drive next-generation vehicles, Innovate energy solutions, Agriculture, Exploring space, Movement of people by luxury cars and passenger drones, Manage the energy transition, Mass transportation, Build intelligent cars and trains.	8	CIA+SEE+Quiz	3,4

Text Books	
1	Marr, Bernard, and Matt Ward. 2019. Artificial Intelligence in Practice : How 50 Successful Companies Used Artificial Intelligence to Solve Problems. Chichester, West Sussex: Wiley. https://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=2105699 .

Reference Books	
1	Deep Learning (Ian J. Goodfellow, Yoshua Bengio and Aaron Courville), MIT Press, 2016.



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SOFTWARE TESTING

Course Code	21AI752	Credits	3
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	OEC	Exam Hours	3

Course Objectives

1	Apply the concepts of software testing to assess the most appropriate testing method.
2	Analyze the importance of testing in software development.
3	Evaluate the suitable testing model to derive test cases for any given software
4	Develop appropriate document for the software artefact.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Apply the concepts of software testing to assess the most appropriate testing method.	L3
CO2	Analyze the importance of testing in software development.	L4
CO3	Evaluate the suitable testing model to derive test cases for any given software	L5
CO4	Develop appropriate document for the software artefact.	L6

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3												2	
CO2		3											2	
CO3				3									2	
CO4										3			2	

Module	Topic	Hours	Assessment	CO
1	Basics of Software Testing: Humans, Errors and Testing, Software Quality, Requirements Behavior and Correctness, Correctness versus Reliability, Testing and Debugging, Test Metrics, Testing and Verification, Test-generation Strategies, Static Testing. A Perspective on Testing: Definitions, Test Cases, Insights from Venn Diagram, Identifying Test Cases, Error and fault taxonomies, Levels of testing. Examples: Generalized pseudocode, the Triangle problem, the NextDate function, the Commission problem, the SATM system, the Currency converter, Saturn windshield wiper	8	CIA,SEE, Quiz	1,2,3
2	Functional Testing: Boundary Value Testing - Boundary value analysis, Robustness testing, Worst-case testing, Special Value Testing, Examples, Random Testing, Guidelines. Equivalence Class Testing - Equivalence classes, Equivalence test cases for the triangle problem, NextDate function, and the commission problem, Guidelines and observations, Decision Table Based Testing - Decision tables, Test cases for the triangle problem, NextDate function, and the commission problem, Guidelines and observations.	8	CIA,SEE, Quiz	2,3



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3	Structural Testing: Overview, Statement testing, Program testing, Condition testing, Path testing - DD paths, Test coverage metrics, Basis path testing, guidelines and observations, Dataflow testing: Definition-Use testing, Slice-based testing, Guidelines and observations.	8	CIA,SEE, Quiz	1,2,3
4	Levels of Testing: Traditional view of testing levels, Alternative life-cycle models, The SATM system, Separating integration and system testing. Integration Testing: A closer look at the SATM system, Decomposition-based, call graph-based, Pathbased integrations.	8	CIA,SEE, Quiz	2,3,4
5	System Testing: Threads, Requirement Specification, Finding Threads, Structural strategies for thread testing, SATM test threads System testing guidelines, ASF testing example. Interaction Testing: Context of interaction, A taxonomy of interactions, Interaction, composition, and determinism, Client/Server Testing	8	CIA,SEE, Quiz	2,3,4

Text Books	
1	Paul C. Jorgensen: Software Testing, A Craftsman's Approach, 3rd Edition, Auerbach Publications, 2008.
2	Aditya P Mathur: Foundations of Software Testing, Pearson Education, 2008.

Reference Books	
1	Mauro Pezze, Michal Young: Software Testing and Analysis – Process, Principles and Techniques, Wiley India, 2009.
2	Software testing Principles and Practices – Gopalaswamy Ramesh, Srinivasan Desikan, 2 nd Edition, Pearson, 2007.
3	Software Testing – Ron Patton, 2nd edition, Pearson Education, 2004. 4. The Craft of Software Testing – Brian Marrick, Pearson Education, 1995. 5. Anirban Basu, Software Quality Assurance, Testing and Metrics, PHI, 2015.



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